

CHAPTER 1:

1.1 DATA REPRESENTATION

1.1. NUMBER SYSTEMS

1.2 TEXT, SOUND AND IMAGES

1.3 DATA STORAGE AND COMPRESSION

By: Nouredine Tadjerout

Learning Objectives

1.1 Number systems

- 1 Understand how and why computers use binary to represent all forms of data
 - 2 (a) Understand the denary, binary and hexadecimal number systems
(b) Convert between
 - (i) positive denary and positive binary
 - (ii) positive denary and positive hexadecimal
 - (iii) positive hexadecimal and positive binary
 - 3 Understand how and why hexadecimal is used as a beneficial method of data representation
 - 4 (a) Add two positive 8-bit binary integers
(b) Understand the concept of overflow and why it occurs in binary addition
- Any form of data needs to be converted to binary to be processed by a computer
 - Data is processed using logic gates and stored in registers
 - Denary is a base 10 system
 - Binary is a base 2 system
 - Hexadecimal is a base 16 system
 - Values used will be integers only
 - Conversions in both directions, e.g. denary to binary or binary to denary
 - Maximum binary number length of 16-bit
 - Areas within computer science that hexadecimal is used should be identified
 - Hexadecimal is easier for humans to understand than binary, as it is a shorter representation of the binary
 - An overflow error will occur if the value is greater than 255 in an 8-bit register
 - A computer or a device has a predefined limit that it can represent or store, for example 16-bit
 - An overflow error occurs when a value outside this limit should be returned

Learning Objectives

1.1 Number systems

- 5 Perform a logical binary shift on a positive 8-bit binary integer and understand the effect this has on the positive binary integer
 - Perform logical left shifts
 - Perform logical right shifts
 - Perform multiple shifts
 - Bits shifted from the end of the register are lost and zeros are shifted in at the opposite end of the register
 - The positive binary integer is multiplied or divided according to the shift performed
 - The most significant bit(s) or least significant bit(s) are lost
- 6 Use two's complement to represent positive and negative 8-bit binary integers
 - Convert a positive binary or denary integer to a two's complement 8-bit integer and vice versa
 - Convert a negative binary or denary integer to a two's complement 8-bit integer and vice versa

Learning Objectives

1.2 Text, sound and images

- 1 Understand how and why a computer represents text and the use of character sets, including American standard code for information interchange (ASCII) and Unicode
 - Text is converted to binary to be processed by a computer
 - Unicode allows for a greater range of characters and symbols than ASCII, including different languages and emojis
 - Unicode requires more bits per character than ASCII
- 2 Understand how and why a computer represents sound, including the effects of the sample rate and sample resolution
 - A sound wave is sampled for sound to be converted to binary, which is processed by a computer
 - The sample rate is the number of samples taken in a second
 - The sample resolution is the number of bits per sample
 - The accuracy of the recording and the file size increases as the sample rate and resolution increase
- 3 Understand how and why a computer represents an image, including the effects of the resolution and colour depth
 - An image is a series of pixels that are converted to binary, which is processed by a computer
 - The resolution is the number of pixels in the image
 - The colour depth is the number of bits used to represent each colour
 - The file size and quality of the image increases as the resolution and colour depth increase

Learning Objectives

1.3 Data storage and compression

1 Understand how data storage is measured

- Including:
 - bit
 - nibble
 - byte
 - kibibyte (KiB)
 - mebibyte (MiB)
 - gibibyte (GiB)
 - tebibyte (TiB)
 - pebibyte (PiB)
 - exbibyte (EiB)
- The amount of the previous denomination present in the data storage size, e.g.:
 - 8 bits in a byte
 - 1024 mebibytes in a gibibyte
- Answers must be given in the units specified in the question
- Information given may include:
 - image resolution and colour depth
 - sound sample rate, resolution and length of track
- Compression exists to reduce the size of the file
- The impact of this is, e.g.:
 - less bandwidth required
 - less storage space required
 - shorter transmission time

2 Calculate the file size of an image file and a sound file, using information given

3 Understand the purpose of and need for data compression

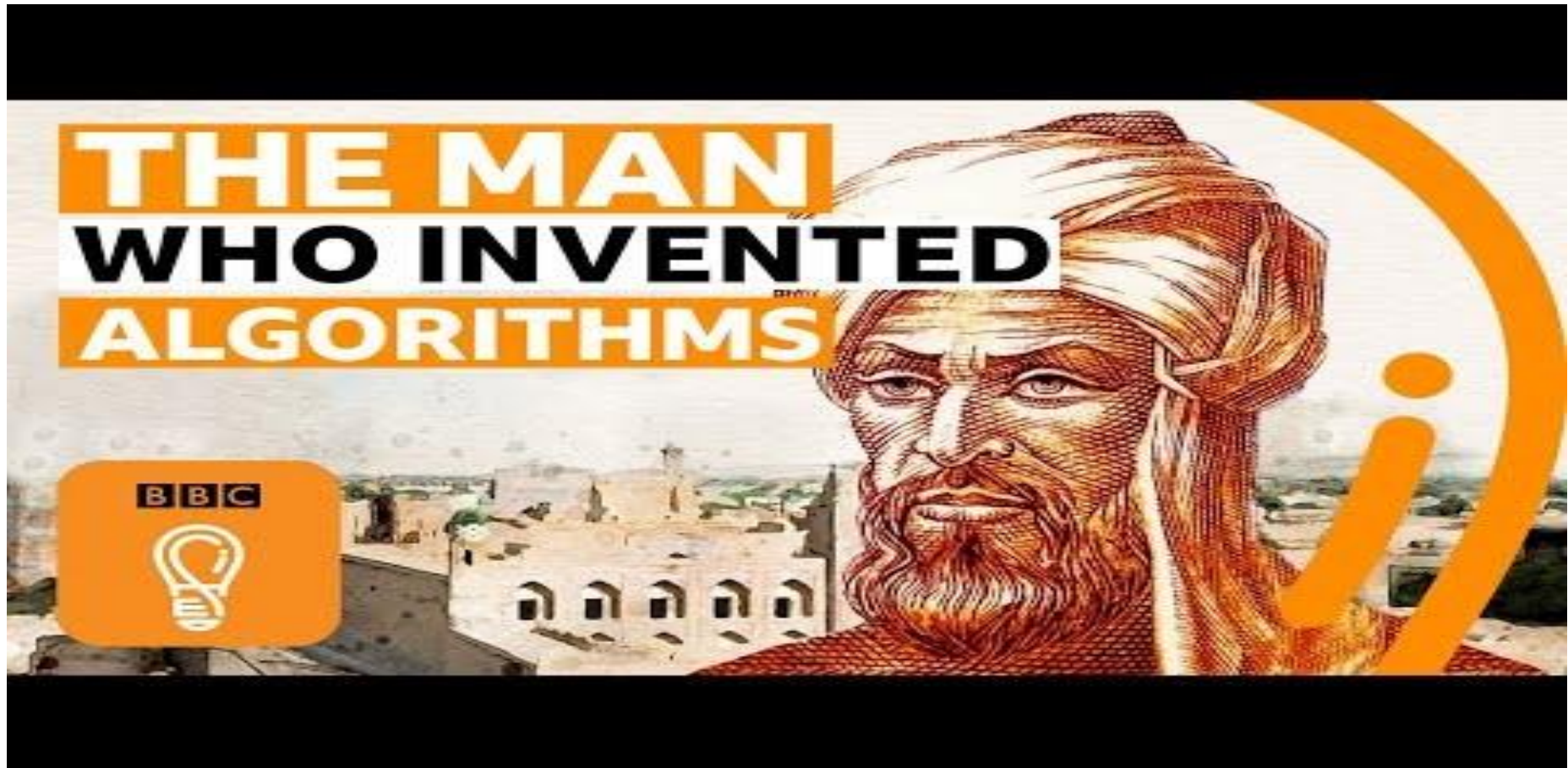
Learning Objectives

1.3 Data storage and compression

- 4 Understand how files are compressed using lossy and lossless compression methods
 - Lossless compression reduces the file size without permanent loss of data, e.g. run length encoding (RLE)
 - Lossy compression reduces the file size by permanently removing data, e.g. reducing resolution or colour depth, reducing sample rate or resolution

Why algorithms are called algorithm

<https://www.youtube.com/watch?v=oRkNaF0QvnI>



Why algorithms are called algorithm

كيف وضع الخوارزمي الأرقام العربية



زاوية واحدة



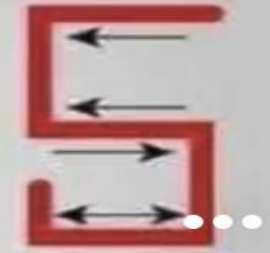
زاويتان



ثلاث زوايا



أربع زوايا



خمس زوايا



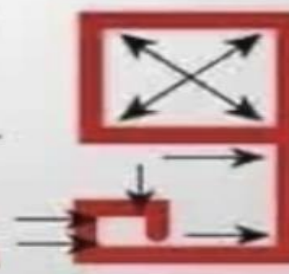
ست زوايا



سبع زوايا



ثمان زوايا



تسع زوايا



بدون زوايا



144K



COMPUTER SCIENCE

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Put these into Size Order – 30secs



Megabyte (Mb)

Terabyte (Tb)

Kilobyte (Kb)

bit

Byte

Gigabyte (Gb)

Rank these types of data storage in size

RAM



4-8Gb



128Gb



700Mb

Hard Drive



4Tb



256Gb



4.7Gb

1.1 Number systems

Measurement of the size of computer memories

Name of memory size	Equivalent denary value (bytes)
1 kilobyte (1 KB)	1 000
1 megabyte (1 MB)	1 000 000
1 gigabyte (1 GB)	1 000 000 000
1 terabyte (1 TB)	1 000 000 000 000
1 petabyte (1 PB)	1 000 000 000 000 000

Name of memory size	Number of bytes	Equivalent denary value (bytes)
1 kibibyte (1 KiB)	2^{10}	1 024
1 mebibyte (1 MiB)	2^{20}	1 048 576
1 gibibyte (1 GiB)	2^{30}	1 073 741 824
1 tebibyte (1 TiB)	2^{40}	1 099 511 627 776
1 pebibyte (1 PiB)	2^{50}	1 125 899 906 842 624

1.1 Number systems

The Difference Between Kilobytes and Kibibytes

Multiples of bytes						V • T • E
Decimal			Binary			
Value		Metric	Value	IEC	JEDEC	
1000	kB	kilobyte	1024	KiB kibibyte	KB	kilobyte
1000 ²	MB	megabyte	1024 ²	MiB mebibyte	MB	megabyte
1000 ³	GB	gigabyte	1024 ³	GiB gibibyte	GB	gigabyte
1000 ⁴	TB	terabyte	1024 ⁴	TiB tebibyte	—	
1000 ⁵	PB	petabyte	1024 ⁵	PiB pebibyte	—	
1000 ⁶	EB	exabyte	1024 ⁶	EiB exbibyte	—	
1000 ⁷	ZB	zettabyte	1024 ⁷	ZiB zebibyte	—	
1000 ⁸	YB	yottabyte	1024 ⁸	YiB yobibyte	—	

This system is more accurate. Internal memories (such as RAM) should be measured using the IEC system. A 64 GiB RAM could, therefore, store 64×2^{30} bytes of data (68 719 476 736 bytes).

1.1 Number systems

Bits, Bytes & Nibbles

Bit – e.g. 0

Each individual 1 or 0 is known as a **bit**.

Here are three bits – 1 1 0

Byte – e.g. 1 1 0 0 1 1 0 0

Each group of 8 bits is known as a **Byte**

Here is a stream of Bytes:

01010011 01101011 01111001 01110010 01101001 01101101 00100000 01101001
01110011 00100000 01100001 01110111 01100101 01110011 01101111 01101101
01100101 00100001

Nibble – e.g. 1 1 1 1

A nibble is 4 bits, or half a Byte.

Here is a nibble – 1 1 0 1

Data Storage Capacities

Terabyte:

A Terabyte is a measure of computer storage capacity and is 2 to the 40th power of 1024 gigabytes.

Petabyte:

A Petabyte (PB) is a measure of memory or storage capacity and is 2 to the 50th power bytes or, in decimal, approximately a thousand terabytes (1024 terabytes).

Exabyte:

An Exabyte (EB) is a large unit of computer data storage, two to the sixtieth power bytes. The prefix exa means one billion, or on quintillion, which is a decimal term. Two to the sixtieth power is actually 1,152,921,504,606,846,976 bytes in decimal, or somewhat over a quintillion (or ten to the eighteenth power) bytes. It is common to say that an Exabyte is approximately one quintillion bytes. In decimal terms, an Exabyte is a billion gigabytes.

Zettabyte:

A Zettabyte (ZB) is equal to one sextillion bytes. It is commonly abbreviated ZB. At this time, no computer has one Zettabyte of storage. It has 1024 Exabytes.

Yottabyte:

A Yottabyte is equal to one septillion bytes. It is commonly abbreviated YB. At this time, no computer has one Zettabyte of storage. It has 1024 Zettabytes.

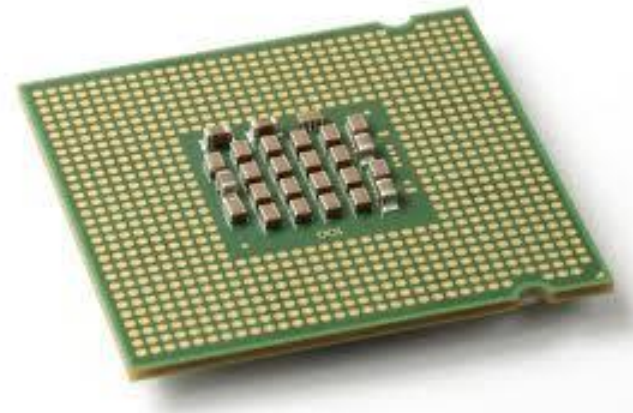
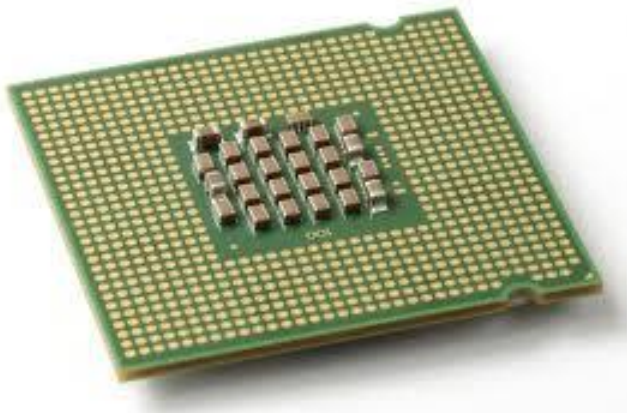
How Computers Work – Binary!

- Computers are made up of complicated hardware that stores and processes data.
- If you break a computer down into its most basic components you have millions of circuits that either allow electricity to flow, or not.
- Imagine a whole row of light switches that you can switch on and off in different combinations to mean different things.
- Each switch is either on or off. It only has two states. That is why everything stored in a computer can be stored as a series of 1s and 0s. This is called **binary**.

CPU / Processor/ Chip

- The CPU is the brain of the computer – it tells the computer what to do and when.
- Has millions of tiny switches that can either be:

on or off
1 or 0

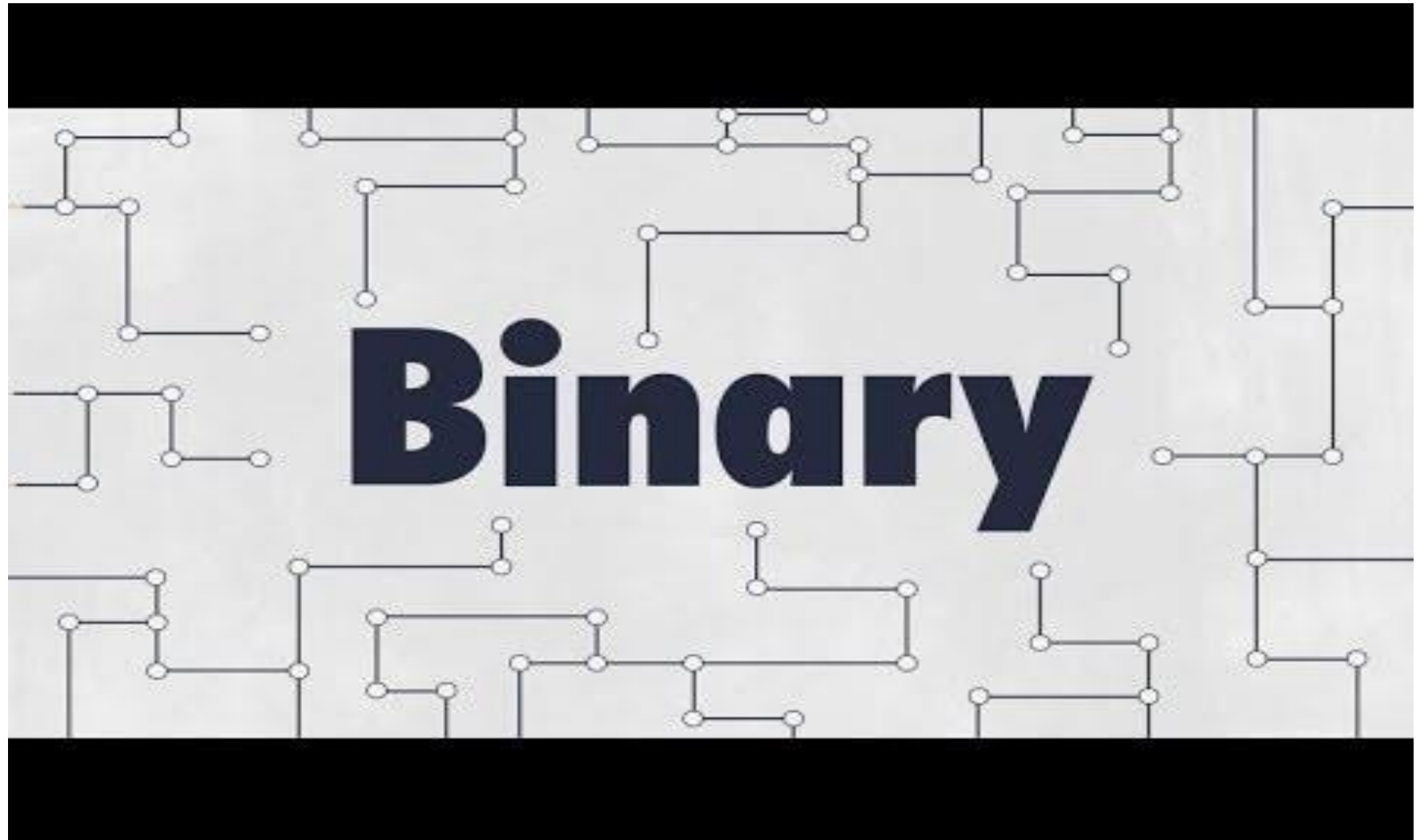


1.1.1.1- Number System (Denary, also known as "decimal")

0 , 1, 2, 3, 4, 5, 6, 7, 8, 9

- We use a number system that uses 10 different numbers. (Base 10 number system)
- This is called the **Denary System**.
- However computers don't understand this as they only understand 0's and 1's.
on or **off**

Click this video that's explains some of the core concepts behind computer science : <https://youtu.be/M41M9ATm49M>



1.1 Number systems

A)-Converting from denary to binary

<https://www.youtube.com/watch?v=rsxT4FfRBaM>

Decimal to Binary

75 → 1001011

142 → 10001110

339 → 101010011

Click the video below to learn to count in binary:

<https://youtu.be/b7pOcU1xMks>

$$(1 \times 8) + (1 \times 4) + (0 \times 2) + (0 \times 1) = 12$$



8



4



2



1

1.1.Number System:

A)-Converting from denary to binary

- The position of numbers in binary is also important.
- Calculating denary number 1:



Denary	128	64	32	16	8	4	2	1
Binary	0	0	0	0	0	0	0	1

$$(1 * 1) = 1$$

Binary = 00000001

1.1. Number System:

A)-Converting from denary to binary

- The position of numbers in binary is also important.
- Calculating denary number 3:

Denary	128	64	32	16	8	4	2	1
Binary	0	0	0	0	0	0	1	1


$$(1 * 2) = 2 + (1 * 1) = 1$$

Binary = 00000011

1.1. Number System:

A)-Converting from denary to binary

- The position of numbers in binary is also important.
- Calculating denary number 6:

Denary	128	64	32	16	8	4	2	1
Binary	0	0	0	0	0	1	1	0


$$(1 * 4) = 4 + (1 * 2) = 2$$

Binary = 00000110

1.1. Number System:

A)-Converting from denary to binary

- The position of numbers in binary is also important.
- Calculating denary number 37:

Denary	128	64	32	16	8	4	2	1
Binary	0	0	1	0	0	1	0	1


$$(1 * 32) = 32 + (1 * 4) = 4 + (1 * 1) = 1$$

Binary = 00100101

1.1. Number System:

A)-Converting from denary to binary

- Can you work out how to write the denary number 115 in binary?

Denary	128	64	32	16	8	4	2	1
Binary								

↑

64

↑

32

↑

16

↑

2

↑

1

+

+

+

+

+

Binary = 01110011

1.1. Number System:

A)-Converting from denary to binary

- Can you work out how to write the denary number 255 in binary?

Denary	128	64	32	16	8	4	2	1							
Binary															
	↑	↑	↑	↑	↑	↑	↑	↑							
	128	+	64	+	32	+	16	+	8	+	4	+	2	+	1

Binary = 11111111

1.1. Number System:

A)-Converting from denary to binary



- Work out the following numbers in binary:

Number	Binary
165	
117	
61	
224	
39	
139	
170	
186	
255	
223	
84	

Extension: Activity 1.1



Convert the following denary numbers into binary (using both methods):

a: 42

b: 69

c: 88

d: 101

e: 112

f: 127

g: 146

h: 189

i: 204

Extension: IGCSE Exam Question 1

Benedict has a computer that is assigned an Internet Protocol (IP) address. The IP address is:

198.167.214.0

The IP address is represented as denary values.

(a) Convert the denary values 167 and 214 from the IP address to 8-bit binary.

167

--	--	--	--	--	--	--	--

214

--	--	--	--	--	--	--	--

Extension: IGCSE Exam Question 2

A stopwatch uses six digits to display hours, minutes and seconds.

The stopwatch is stopped at:

0	2	:	3	1	:	5	8
Hours			Minutes			Seconds	

An 8-bit register is used to store each pair of digits.

(a) Write the 8-bit binary numbers that are currently stored for the **Hours**, **Minutes** and **Seconds**.

Hours

--	--	--	--	--	--	--	--

Minutes

--	--	--	--	--	--	--	--

Seconds

--	--	--	--	--	--	--	--

IGCSE Exam Question 3

A school network has several computers.

Each computer in the network has a media access control (MAC) address.

Hexadecimal is used for MAC addresses.

Part of a MAC address is given.

97–5C–E1

Each pair of digits is stored as binary in an 8-bit register.

(a) Complete the binary register for these two pairs of digits.

97							
5C							

[4]

(b) Describe what is meant by a MAC address.

.....

IGCSE Exam Answer 3

Question	Answer	Marks																
1(a)	One mark per each correct four bits. 97 <table border="1"><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr></table> 5C <table border="1"><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table>	1	0	0	1	0	1	1	1	0	1	0	1	1	1	0	0	4
1	0	0	1	0	1	1	1											
0	1	0	1	1	1	0	0											
1(b)	Any four from: • used to identify a device • it is a <u>unique</u> (address) • it is a static address // it does not change • it is set by the manufacturer • the first part is the manufacturer ID / number / identifies the manufacturer • the second part is the serial number / ID.	4																

1.1. Number System:

B)-Converting from binary to denary or Decimal

<https://www.youtube.com/watch?v=VLfITjd3IWA>

Binary to Decimal

1 1 0 0 0 1 1 $\rightarrow$ 99

1 1 0 0 . 1 0 1 $\rightarrow$ 12.625

Tip – Always use 8 bits for binary!

- So far we have only looked at numbers using eight columns. This is an 8-bit number, or a byte.
- **00000011 is binary for 3.**
- You do not need the leading zeros for it to be a valid number but we tend to write groups of 8 bits because computers usually store data in bytes.

1.1. Number System:

B)-Converting from binary to denary or Decimal

Convert the following binary numbers into denary:

a: 0 0 1 1 0 0 1 1

b: 0 1 1 1 1 1 1 1

c: 1 0 0 1 1 0 0 1

d: 0 1 1 1 0 1 0 0

e: 1 1 1 1 1 1 1 1

f: 0 0 0 0 1 1 1 1

g: 1 0 0 0 1 1 1 1

h: 1 1 1 1 0 0 0 0

i: 0 1 1 1 0 0 0 0

j: 1 1 1 0 1 1 1 0

IGCSE Exam Question 1

(b) The stopwatch is started again and then stopped.

When the watch is stopped, the 8-bit binary registers show:

Hours	0	0	0	0	0	1	0	1
Minutes	0	0	0	1	1	0	1	0
Seconds	0	0	1	1	0	1	1	1

Write the denary values that will now be shown on the stopwatch.

	:	:
	:	:
Hours	Minutes	Seconds

Review



- What is binary? Explain this as if you are explaining this to someone who knows nothing about it.

1.1 Number systems

Measurement of the size of computer memories

The Difference Between Kilobytes and Kibibytes

A kibibyte is the formal name for 1024 Bytes. It is “supposed” to be used instead of kilobyte for this quantity, which according to SI naming conventions is supposed to be used for 1000 bytes. Kibibyte, Kilobyte.

In my experience, the “kibi” terms seem to be rarely used, as just about every computer person other than some people on the net uses kilobyte to mean 1024 bytes and megabyte (or “meg”) to be 1024^2 bytes (versus “mebibyte”, which I’ve seen even more rarely).

In practice, powers of 1024 bytes tend to end up being used for RAM, while powers of 1000 end up being used for selling hard disks and other persistent storage. Confusingly, file sizes in bytes are also typically reported in terms of 1024 byte powers by most Unix/Linux file size tools, such as “ls -lh”. (ls -lh. This command will show you the file sizes in human readable format) Whenever I hear the word “kibibyte”, I think of this...



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1000 ⁴	TB	terabyte	1024 ⁴	TiB tebibyte	—	
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This system is more accurate. Internal memories (such as RAM) should be measured using the IEC system. A 64 GiB RAM could, therefore, store 64×2^{30} bytes of data (68 719 476 736 bytes).

1.1 Data Representation

Number systems: Binary addition and subtraction

Click the link below to watch the video to How To Add and Subtract Binary Numbers:

<https://www.youtube.com/watch?v=C5EkxfNEMjE>

Binary Operations

$$\begin{array}{r} 100110 \\ + 110101 \\ \hline 1011011 \end{array} \quad \begin{array}{r} 100 \\ - 1 \\ \hline 011 \end{array}$$

Number systems: Addition of binary numbers:

We will look at the addition of two 8-bit positive binary numbers.

Note the following key facts when carrying out addition of two binary digits:

binary addition	carry	sum
0+0	0	0
0+1	0	1
1+0	0	1
1+1	1	0

? Example 1

Add 00100111 + 01001010

We will set this out showing **carry** and **sum** values:

$$\begin{array}{r} 00100111 \\ + 01001010 \\ \hline 111 \\ \hline 01110001 \end{array}$$

← carry values

← sum values

Answer: 01110001

column 1: $1 + 0 = 1$ no carry
column 2: $1 + 1 = 0$ carry 1
column 3: $1 + 0 + 1 = 0$ carry 1
column 4: $0 + 1 + 1 = 0$ carry 1
column 5: $0 + 0 + 1 = 1$ no carry
column 6: $1 + 0 = 1$ no carry
column 7: $0 + 1 = 1$ no carry
column 8: $0 + 0 = 0$ no carry

Number systems: Addition of binary numbers:

? Example 2

a Convert 126 and 62 into binary.

b Add the two binary values in part **a** and check the result matches the addition of the two denary numbers

a $126 = 01111110$ and $62 = 00111110$

b 01111110

+

00111110

111111

← carry values

10111100

← sum values

Answer: 10111100

column 1: $0 + 0 = 0$ no carry
column 2: $1 + 1 = 0$ carry 1
column 3: $1 + 1 + 1 = 1$ carry 1
column 4: $1 + 1 + 1 = 1$ carry 1
column 5: $1 + 1 + 1 = 1$ carry 1
column 6: $1 + 1 + 1 = 1$ carry 1
column 7: $1 + 0 + 1 = 0$ carry 1
column 8: $0 + 0 + 1 = 1$ no carry

10111100 has the equivalent denary value of $128 + 32 + 16 + 8 + 4 = 188$ which is the same as $126 + 62$.

Number systems: Addition of binary numbers:

Question 1:

Carry out the following binary additions:

a $00011101 + 01100110$

b $00100111 + 00111111$

c $00101110 + 01001101$

d $01110111 + 00111111$

e $00111100 + 00110011$

Number systems: Binary addition and subtraction:

Question 2:

Add the following binary numbers:

$$\begin{array}{r} 10010 \\ + 1100 \\ \hline \end{array}$$

$$\begin{array}{r} 1011101 \\ + 1000000 \\ \hline \end{array}$$

$$\begin{array}{r} 10011 \\ + 1111101 \\ \hline \end{array}$$

$$\begin{array}{r} 10011001 \\ + 100111 \\ \hline \end{array}$$

$$\begin{array}{r} 11000011 \\ + 101111 \\ \hline \end{array}$$

$$\begin{array}{r} 1001100 \\ + 1100101 \\ \hline \end{array}$$

1.1 Data Representation

Number systems: Binary addition and subtraction

Answer:

$$\begin{array}{r} 10010 \\ + 1100 \\ \hline 11110 \end{array}$$

$$\begin{array}{r} 1011101 \\ + 1000000 \\ \hline 10011101 \end{array}$$

$$\begin{array}{r} 10011 \\ + 1111101 \\ \hline 10010000 \end{array}$$

$$\begin{array}{r} 10011001 \\ + 100111 \\ \hline 11000000 \end{array}$$

$$\begin{array}{r} 11000011 \\ + 101111 \\ \hline 11110010 \end{array}$$

$$\begin{array}{r} 1001100 \\ + 1100101 \\ \hline 10110001 \end{array}$$

1.1 Data Representation

Number systems: Logical binary shifts

Computers can carry out a **logical shift** on a sequence of binary numbers. The logical shift means moving the binary number to the *left* or to the *right*. Each shift *left* is equivalent to ***multiplying*** the binary number by 2 and each shift *right* is equivalent to ***dividing*** the binary number by 2.

? Example 1

The denary number 21 is 00010101 in binary. If we put this into an 8-bit register:

128	64	32	16	8	4	2	1
0	0	0	1	0	1	0	1

The left-most bit is often referred to as the MOST SIGNIFICANT BIT

If we now shift the bits in this register one place to the left, we obtain:

128	64	32	16	8	4	2	1
0	0	1	0	1	0	1	0

The left-most bit is now lost following a left shift

Note how the empty right-most bit position is now filled with a 0

The value of the binary bits is now 21×2^1 i.e. 42. We can see this is correct if we calculate the denary value of the new binary number 101010 (i.e. $32 + 8 + 2$).

1.1 Data Representation

Number systems: Logical binary shift: Left shift

Suppose we now shift the original number two places left:

128	64	32	16	8	4	2	1
0	1	0	1	0	1	0	0

The binary number 1010100 is 84 in denary – this is 21×2^2 .

And now suppose we shift the original number three places left:

1	0	1	0	1	0	0	0
---	---	---	---	---	---	---	---

The binary number 10101000 is 168 in denary – this is 21×2^3 .

1.1 Data Representation

Number systems: Logical binary shift: Right shift

? Example 2

The denary number 200 is 11001000 in binary. Putting this into an 8-bit register gives:

128	64	32	16	8	4	2	1
1	1	0	0	1	0	0	0

The right-most bit is often referred to as the LEAST SIGNIFICANT BIT

If we now shift the bits in this register one place to the right:

128	64	32	16	8	4	2	1
0	1	1	0	0	1	0	0

Note how the left-most bit position is now filled with a 0

1.1 Data Representation

Number systems: Logical binary shift: Left shift

The value of the binary bits is now $200 \div 2^1$ i.e. 100. We can see this is correct by converting the new binary number 01100100 to denary ($64 + 32 + 4$).

Suppose we now shift the original number two places to the right:

128	64	32	16	8	4	2	1
0	0	1	1	0	0	1	0

The binary number 00110010 is 50 in denary – this is $200 \div 2^2$.

And suppose we now shift the original number three places to the right:

128	64	32	16	8	4	2	1
0	0	0	1	1	0	0	1

Notice the 1-bit from the right-most bit position is now lost causing an error

The binary number 00011001 is 25 in denary – this is $200 \div 2^3$.

Now let us consider what happens if we shift four places right:

128	64	32	16	8	4	2	1
0	0	0	0	1	1	0	0

The right-most 1-bit has been lost. In our 8-bit register the result of $200 \div 2^4$ is 12, which is clearly incorrect. This error is because we have therefore exceeded the maximum number of right shifts possible using this 8-bit register.

1.1 Data Representation

Number systems: Logical binary shift: Left shift

? Example 3

- a Write 24 as an 8-bit register.
- b Show the result of a logical shift 2 places to the left.
- c Show the result of a logical shift 3 places to the right.

a

128	64	32	16	8	4	2	1
0	0	0	1	1	0	0	0

b

128	64	32	16	8	4	2	1
0	1	1	0	0	0	0	0

↔ $24 \times 2^2 = 96$

c

128	64	32	16	8	4	2	1
0	0	0	0	0	0	1	1

↔ $24 \div 2^3 = 3$

1.1 Data Representation

Number systems: Logical binary shift: Left shift

- 1 a Write down the denary value of the following binary number.

0	1	1	0	1	0	0	0
---	---	---	---	---	---	---	---

- b Shift the binary number three places to the right and comment on your result.

- c Write down the denary value of the following binary number.

0	0	0	0	1	1	1	1
---	---	---	---	---	---	---	---

- d Shift the binary number four places to the left and comment on your result.

- 2 a Convert 29 and 51 to 8-bit binary numbers.

- b Add the two binary numbers in part a.

- c Shift the result in part b three places to the right.

- d Convert 75 to an 8-bit binary number.

- e Add the two binary numbers from parts c and d.

- f Shift your result from part e one place to the left.

COMPUTER SCIENCE

CHAPTER 1:

1.1 DATA REPRESENTATION

1.1.2 Hexadecimal

By: Nouredine Tadjerout

Recap



- So if binary is a base 2 number system that has the numbers 0 and 1.
- And denary (base 10) uses the numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.
- How can we display numbers bigger than 9? For example, 10, 11, 12, 13 etc?

We'll come back to this...

Why hexadecimal?

- Hex numbers are shorter/more memorable than equivalent binary numbers..
- ... and can easily be converted to and from binary...
- ... as each hexadecimal digit corresponds to 4 binary digits

Which one is easier to remember?

01011011 OR 5B

- Humans are not very good at remembering long strings of numbers.
- To make it easier for us, we can represent every group of 4 bits (a nibble) with a single digit.



Hexadecimal

- Hexadecimal (or **hex**) is a **base 16** system used to simplify how **binary** is represented. A hex digit can be any of the following 16 digits: **0 1 2 3 4 5 6 7 8 9 A B C D E F**.
- Each hex digit reflects a **4-bit** binary sequence.
- This table below shows each hex digit with the equivalent values in binary and denary.

Hexadecimal

- Hexadecimal is a base 16 number system.

Denary	Hexadecimal
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9

Denary	Hexadecimal
10	A
11	B
12	C
13	D
14	E
15	F

Hexadecimal is a base 16 number system

Denary	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Hexadecimal

- This means an 8-bit binary number can be written using only two different hex digits - one hex digit for each nibble (or group of 4-bits). It is much easier to write numbers as hex than to write them as binary numbers.
- For example:
- **11010100** in binary would be **D4** in hex
- **FFFF3** in hex would be **11111111111111110011** in binary

Using hexadecimal

- **Hex** codes are used in many areas of computing to simplify **binary** codes. It is important to note that computers do not use hexadecimal - it is used by humans to shorten binary to a more easily understandable form. Hexadecimal is translated into binary for computer use. Some examples of where hex is used include:
 - colour references
 - **assembly language** programs
 - error messages

Using hexadecimal: Colours

- Hex can be used to represent **colours** on web pages and image-editing programs using the format **#RRGGBB** (RR = reds, GG = greens, BB = blues). The # symbol indicates that the number has been written in hex format.
- This system uses two hex digits for each colour, eg **#FF6600**.

Using hexadecimal: Colours

- Hex can be used to represent **colours** on web pages and image-editing programs using the format **#RRGGBB** (RR = reds, GG = greens, BB = blues). The # symbol indicates that the number has been written in hex format.
- This system uses two hex digits for each colour, eg #FF6600.



1	1	1	1	=	F	R
1	1	1	1	=	F	
0	1	1	0	=	6	G
0	1	1	0	=	6	
0	0	0	0	=	0	B
0	0	0	0	=	0	

Using hexadecimal: Colours

- As one hex digit represents 4 **bits**, two hex digits together make 8 bits (1 **byte**). The values for each colour run between 00 and FF. In binary, 00 is 0000 0000 and FF is 1111 1111. That provides 256 possible values for each of the three colours.
- That gives a total **spectrum** of 256 reds x 256 greens x 256 blues - which is over 16 million colours in total.
- **#FF0000** will be the purest red - red only, no green or blue.
- Black is **#000000** - no red, no green and no blue.
- White is **#FFFFFF**.

spectrum mean is a band of colours

Using hexadecimal: Colours

- An orange colour can be represented by the code `#FF6600`. The hex code is much easier to read than the binary equivalent 1111 1111 0110 0110 0000 0000.
- If you are making a web page with **HTML** or **CSS** you can use hex codes to choose the colours.

RGB colour model

- Hex values have equivalents in the **RGB colour model**. The RGB model is very similar to the hex colour model, but instead of combining hex values you use a value between 0 and 255 for each colour. So an orange colour that is `#FF6600` in hex would be 255, 102, 0 in RGB.

Memory dumps

- Since it is much easier to work with: **B 5 A 4 1 A F C**
- Rather than: 1 0 1 1 | 1 0 0 1 | 1 0 1 0 | 0 1 0 0 | 0 0 0 1 | 1 0 1 0 | 1 1 1 1 | 1 1 0 0
- Hexadecimal is often used when developing new software or when trying to trace errors in programs. The contents of part of the computer memory can hold the key to help solve many problems. When the memory contents are output to a printer or monitor, this is known as a **MEMORY DUMP**.

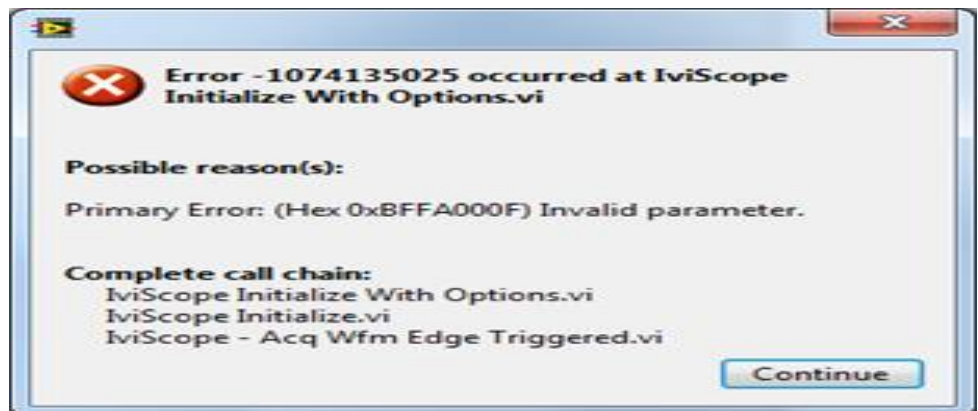
00990F60	54	68	69	73	20	69	73	20	61	6E	20	65	78	61	6D	70	6C	65	20	6F	66
00990F77	61	20	6D	65	6D	6F	72	79	20	64	75	6D	70	20	66	72	6F	6D	20	20	61
00990E8E	74	79	70	69	63	61	6C	20	20	63	6F	6D	70	75	74	65	72	20	20	6D	85
00990EA5	6D	6F	72	79	20	73	68	6F	77	69	6E	67	20	74	68	65	20	20	63	6F	6E
00990EBC	74	65	6E	74	73	20	6F	66	20	61	20	6E	75	6D	62	65	72	20	20	6F	66
00990ED3	6C	6F	63	61	74	69	6F	6E	73	20	20	69	6E	20	20	68	65	78	20	20	20
00990EEA	6E	6F	74	61	74	69	6F	6E	20	20	00	00	00	00	00	00	00	00	00	00	00

Figure 1.1.2.1

Memory dumps

A program developer can look at each of the hexadecimal codes (as shown in Figure 1.1.2.1) and determine where the error lies. The value on the far left shows the memory location so that it is possible to find out exactly where in memory the fault occurs.

This is clearly much more manageable using hexadecimal rather than using binary. It's a very powerful fault-tracing tool, but requires considerable knowledge of computer architecture in order to interpret the results.



Memory dumps

Click this video that's explains What is a memory dump?

<https://www.youtube.com/watch?v=bTOElvZnAVQ>

A problem has been detected and windows has been shut down to prevent damage to your computer.

The end-user manually generated the crashdump.

If this is the first time you've seen this stop error screen, read the instructions below. If this screen appears again, follow these steps:

Check to make sure any new hardware or software is properly installed. If this is a new installation, ask your hardware or software manufacturer for any windows updates you might need.

If problems persist, disable or remove any newly installed hardware or software. Disable memory options such as caching or shadowing. If you need to disable memory options, select Advanced Startup Options, and then select Troubleshoot.

Technical information:

*** STOP: 0x000000E2 (0x00000000, 0x00000000, 0x00000000, 0x00000000)

Memory dumps

Click this video How to Remove System Error Memory Dump Files on Windows 10?

<https://www.youtube.com/watch?v=JuWpa0ebyZc>



Using hexadecimal: Errors

- Hex is often used in error messages on your computer. The hex number refers to the **memory location** of the error. This helps programmers to find and then fix problems.

1.2 - Hexadecimal:

- Using the same method as denary and binary, this gives the headings of 16^0 , 16^1 , 16^2 , 16^3 and so on. The typical headings for a hexadecimal number with five digits would be:

65 536	4 096	256	16	1
(16^4)	(16^3)	(16^2)	(16^1)	(16^0)

1.1. Data Representation:

A)-Converting from Hexadecimal to Decimal or (Denary)

<https://www.youtube.com/watch?v=pg-HEGBpCQk>

Hexadecimal to Decimal

23E₁₆

→

574₁₀

↓ ↓ ↓
16² 16¹ 16⁰

A → 10

C → 12

B → 11

D → 13

AB09₁₆

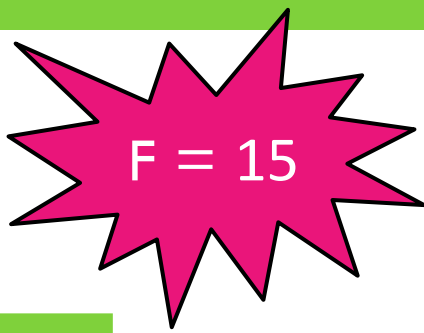
→

43875₁₀

1.1. Data Representation

A)-Converting from Hexadecimal to Decimal or (Denary)

- Hexadecimal Number FF to denary



F = 15

Decimal	4096	256	16	1
Hexadecimal			F	F

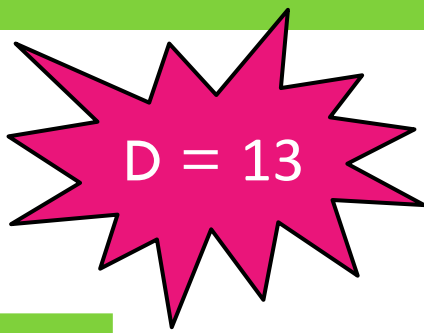

$$((16 * 15) = 240) + ((1 * 15) = 15))$$
$$(240 + 15 = 255)$$

Denary = 255

1.1. Data Representation

A)-Converting from Hexadecimal to Decimal or (Denary)

- Hexadecimal Number 8D to denary



D = 13

Decimal	4096	256	16	1
Hexadecimal			8	D


$$((16 * 8) = 128) + ((1 * 13) = 13)$$

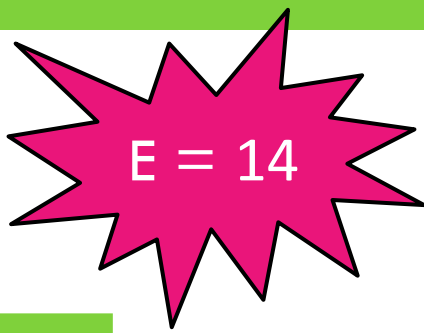
$$((128 + 13 = 141))$$

Denary = 141

1.1. Data Representation

A)-Converting from Hexadecimal to Decimal or (Denary)

- On whiteboards convert EE to denary



E = 14

Decimal	4096	256	16	1
Hexadecimal			E	E


$$(16 * 14) = 224 + (1 * 14) = 14$$

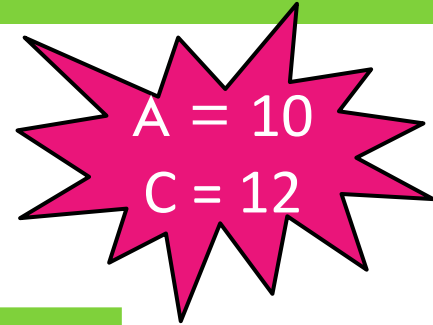
$$(224 + 14) = 238$$

Denary = 238

1.1. Data Representation

A)-Converting from Hexadecimal to Decimal or (Denary)

- Convert 55AC to denary



Decimal	4096	256	16	1
Hexadecimal	5	5	A	C

$$(4096 * 5) + (256 * 5) + (16 * 10) + (1 * 12)$$

Denary = 21932

1.1. Data Representation

A)-Converting from Hexadecimal to Decimal or (Denary

Example 1

4 5 A

First multiply each digit by its value:

$$\begin{array}{llll} 256 & 16 & 1 & \\ (4 \times 256 = 1024) & (5 \times 16 = 80) & (10 \times 1 = 10) & (\text{Note: } A = 10) \end{array}$$

Add the totals together:

denary number = 1114

1.1. Data Representation

A)-Converting from Hexadecimal to Decimal or (Denary

Example 2

C 8 F

First multiply each digit by its value:

$$\begin{array}{ccc} 256 & 16 & 1 \\ (12 \times 256 = 3072) & (8 \times 16 = 128) & (15 \times 1 = 15) \end{array} \quad (\text{Note: C} = 12 \text{ and F} = 15)$$

Add the totals together:

$$\text{denary number} = 3215$$

1.1. Data Representation

A)-Converting from Hexadecimal to Decimal or (Denary)

Convert the following hexadecimal numbers into denary

a 6 B

b 9 C

c 4 A

d F F

e 1 F F

f A 0 1

g B B 4

h C A 8

i 1 2 A E

j A D 8 9

IGCSE Exam – Question 1

Pradeep is reading hexadecimal values for a project he is working on.

- (a) The first three hexadecimal values he reads are **15**, **102** and **A9**.

Give the **denary** values for the three hexadecimal values.

15

102

A9 [3]

Convert the **two** given hexadecimal values into denary.

29

C8 [2]

1.1. Data Representation

B)-Converting from Decimal To Hexadecimal

<https://www.youtube.com/watch?v=QJW6qnfhC70>

Decimal to Hexadecimal

479₁₀

→

1DF₁₆

15 → F

13 → D

1 → 1



894₁₀

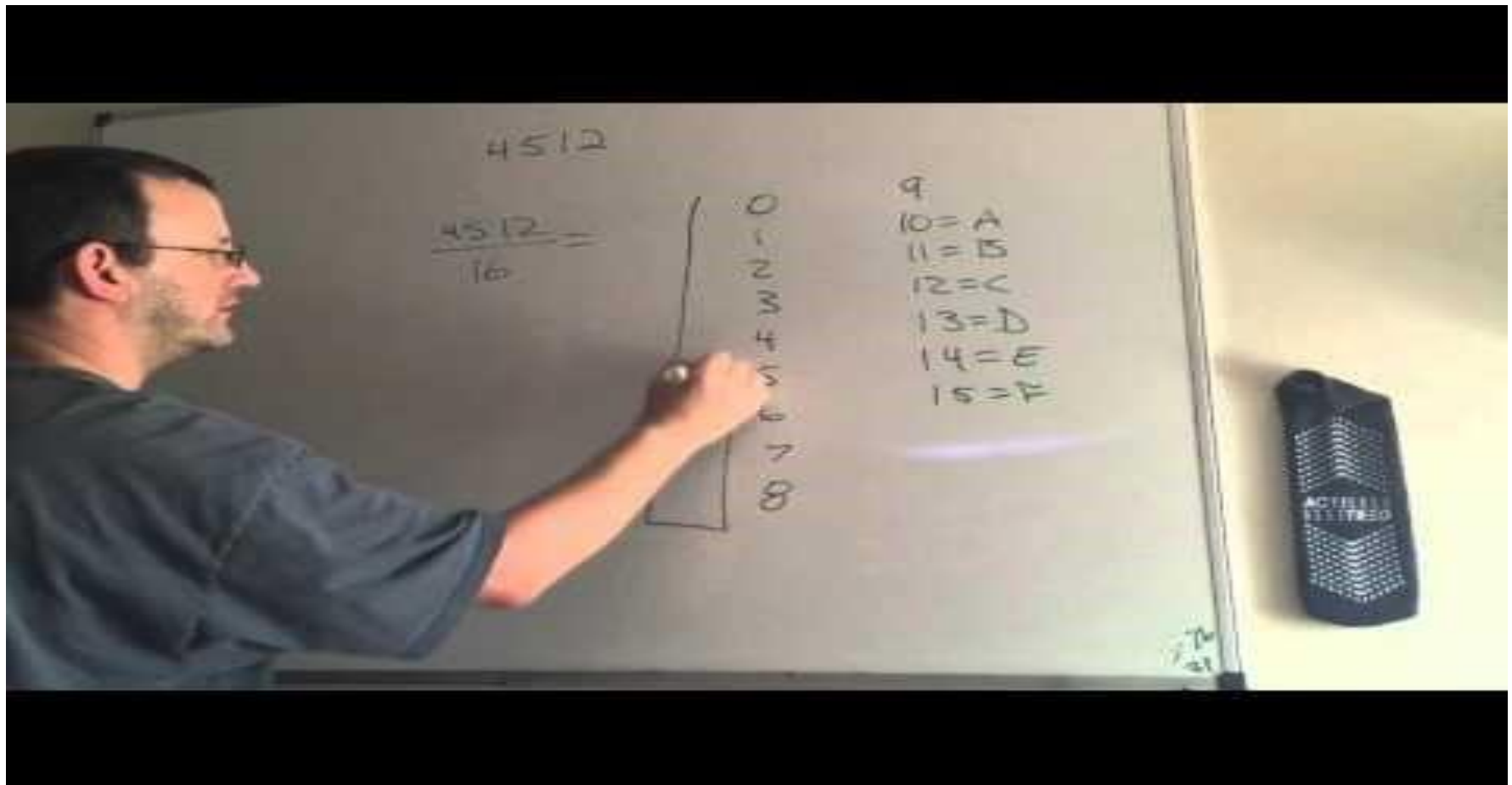
↓

37E₁₆

1.1. Data Representation

B)-Converting from Decimal To Hexadecimal

<https://www.youtube.com/watch?v=4um7-QbpAQ8>



Hexadecimal is a base 16 number system

Denary	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

1.1. Data Representation

B)-Converting from Decimal or Denary To Hexadecimal

Method 1: Convert the denary to binary first and then convert the binary to hex as below:

Consider the conversion of the denary number, 2004, into hexadecimal.

Denary	1024	512	256	128	64	32	16	8	4	2	1
Binary	1	1	1	1	1	0	1	0	1	0	0
	7			D				4			

Now look at Hexadecimal is a base 16 number table system above: you need to start from the left to the right by dividing

01111 = 7 and 1101 = D and 0100 = 4

denary number = 2004, into hexadecimal = 7D4

1.1. Data Representation

B)-Converting from Decimal or Denary To Hexadecimal

□ To verify our finding then we use the method 2 below :

Consider the conversion of the denary number, 2004, into hexadecimal. This method involves placing hexadecimal digits in the appropriate position so that the total

256	16	1	
7	D	4	(Note: D = 13)

A quick check shows that: $(7 \times 256) + (13 \times 16) + (4 \times 1)$ gives 2004.

Converting from hexadecimal to denary and from denary to hexadecimal

Convert the following denary numbers into hexadecimal (using both methods):

a 9 8

b 2 2 7

c 4 9 0

d 5 1 1

e 8 2 6

f 1 0 0 0

g 2 6 3 4

h 3 7 4 3

i 4 0 0 7

j 5 0 0 0

IGCSE Exam Question 1

A denary value can be converted into hexadecimal and binary.

- (a) Complete the table to show the hexadecimal and 8-bit binary values of the given denary values.

Denary	Hexadecimal	8-bit binary
49		
123		
200		

[6]

IGCSE Exam Question 2

The values in the MAC address are hexadecimal values.

Convert the **three** given hexadecimal values into 8-bit binary.

14

A0

C9

[3]

1.1. Data Representation

D)-Converting from Binary to Hexadecimal

<https://www.youtube.com/watch?v=tSLKOKGQq0Y>

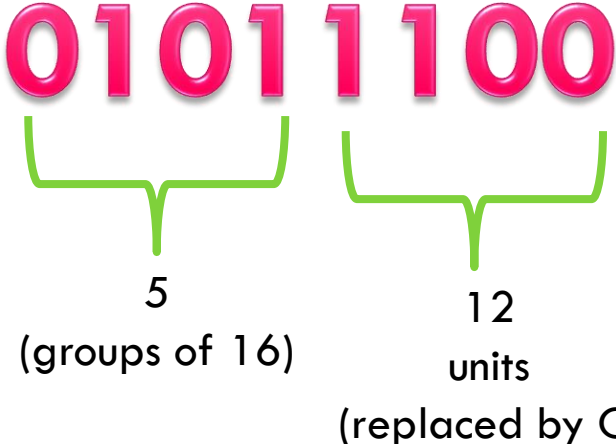
Binary to Hexadecimal

1011 0111 $\rightarrow$ B7₁₆
↓ ↓
11 $\rightarrow$ B 7

10111010 $\rightarrow$ 2EA₁₆

1.1. Data Representation

D)-Converting from Binary to Hexadecimal

- Really you will only ever see 8-bit binary numbers at IGCSE level, which can be represented as two hex digits.
- The first hex digit represents groups of 16, the second hex digit represents the units left over.
- **92** denary =  = **5C** in hex

1.1. Data Representation

D)-Converting from Binary to Hexadecimal

Activity D

Convert the following binary numbers into hexadecimal:

a 1 1 0 0 0 0 1 1

b 1 1 1 1 0 1 1 1

c 1 0 0 1 1 1 1 1 1

d 1 0 0 1 1 1 0 1 1 1 0

e 0 0 0 1 1 1 1 0 0 0 0 1

f 1 0 0 0 1 0 0 1 1 1 1 0

g 0 0 1 0 0 1 1 1 1 1 1 1 0

h 0 1 1 1 0 1 0 0 1 1 1 0 0

i 1 1 1 1 1 1 1 1 0 1 1 1 1 1 0 1

j 0 0 1 1 0 0 1 1 1 1 0 1 0 1 1 1 0

1.1. Data Representation

E)-Converting from Hexadecimal to Binary

https://www.youtube.com/watch?v=D_YC6DSPpQE

Hexadecimal to Binary

A 9 → Binary

↓

1010 1001

A → 10
B → 11
C → 12
D → 13
E → 14
F → 15

1.1. Data Representation

E)-Converting from Hexadecimal to Binary

Activity E

Convert the following hexadecimal numbers into binary:

a 6 C

b 5 9

c A A

d A 0 0

e 4 0 E

f B A 6

g 9 C C

h 4 0 A A

i D A 4 7

j 1 A B 0

1.2 - Hexadecimal:

A)-Hexadecimal Addition

Click the link below to watch the video on

<https://www.youtube.com/watch?v=y0B6tuC6niE>

Hexadecimal Addition

0
1
2
3
4
5
6
7
8
9
A
B
C
D
E
F

10
11
12
13
14
15

i)

$$\begin{array}{r} 5 \ 6 \ 8 \ 9 \\ + \ 4 \ 5 \ 7 \ 4 \\ \hline 9 \ B \ F \ D \end{array}$$

Ans

ii)

$$\begin{array}{r} 16^2 \quad 16^1 \quad 16^0 \\ + \quad \downarrow \quad \downarrow \quad \downarrow \\ A \ D \ D \\ + \ D \ A \ D \end{array}$$

Homework

Hom i)

$$\begin{array}{r} 8 \ 9 \ 9 \\ + \ 1 \ 8 \ 9 \\ \hline \end{array}$$

ii)

$$\begin{array}{r} D \ A \ F \\ + \ B \ A \ F \\ \hline \end{array}$$

47

Digital Electronics

Hexadecimal is a base 16 number system

Denary	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

1.2 - Hexadecimal:

A)-Hexadecimal Addition

Example 1:

★ What hexadecimal value is obtained, when two hex digits C and D are added together?

$$C = 12 = 1100$$

$$D = 13 = 1111$$

$$= \begin{array}{r} 12 \\ 13 \\ \hline 25 \end{array}$$

128	64	32	16	8	4	2	1
0	0	0	1	1	0	0	1
1				9			

$$C + D = 19.$$

1.2 - Hexadecimal:

A)-Hexadecimal Addition

To double check your answer click in the link below:

<https://www.calculator.net/hex-calculator.html>

Hex Calculator

Hexadecimal Calculation—Add, Subtract, Multiply, or Divide

Result

Hex value:

c + D = 19

Decimal value:

12 + 13 = 25

+ = ?

Calculate



Clear

1.2 - Hexadecimal:

A)-Hexadecimal Addition

Example 2: **Hex value:** 8B + AA = 1 3 5

Hex value: 8B = 1000 1011 and **Decimal value:** 139

Denary	1024	512	256	128	64	32	16	8	4	2	1
Binary	0	0	0	1	0	0	0	1	0	1	1

Hex value: AA = 1010 1010 and **Decimal value:** 170

Denary	1024	512	256	128	64	32	16	8	4	2	1
Binary	0	0	0	1	0	1	0	1	0	1	0

Hex value: 8B + AA = 1 3 5 and **Decimal value:** 139 + 170 = 309

Denary	1024	512	256	128	64	32	16	8	4	2	1
Binary	0	0	1	0	0	1	1	0	1	0	1

1

3

5

1.2 - Hexadecimal:

A)-Hexadecimal Addition

To double check your answer click in the link below:

<https://www.calculator.net/hex-calculator.html>

Example 2: Hex Calculator

Hexadecimal Calculation—Add, Subtract, Multiply, or Divide

Result

Hex value:

8B + AA = **135**

Decimal value:

139 + 170 = **309**

+ = ?

Calculate 

Clear

1.2 - Hexadecimal:

A)-Hexadecimal Addition

Activity 1: Hexadecimal Calculation addition:

Hex value:

$$A + E = ?$$

$$D + A = ?$$

$$E7 + AB = ?$$

$$9F + CB = ?$$

$$A5 + EF = ?$$

$$99 + EE = ?$$

$$8B + AA = ?$$

$$B2 + C7 = ?$$

$$A9 + E7 = ?$$

$$AC + EF = ?$$

$$2C + D4 = ?$$

1.2 - Hexadecimal:

B)-Hexadecimal Subtraction

Example 1: **Hex value:** $8B - 7A = 011$
Hex value: $8B = 1000\ 1011$ and **Decimal value:** 139

Denary	1024	512	256	128	64	32	16	8	4	2	1
Binary	0	0	0	1	0	0	0	1	0	1	1

Hex value: $7A = 0111\ 1010$ and **Decimal value:** 122

Denary	1024	512	256	128	64	32	16	8	4	2	1
Binary	0	0	0	0	1	1	1	1	0	1	0

Hex value: $8B - 7A = 11$ and **Decimal value:** $139 - 122 = 17$

Denary	1024	512	256	128	64	32	16	8	4	2	1
Binary	0	0	0	0	0	0	1	0	0	0	1
	0			1			1				

1.2 - Hexadecimal:

B)-Hexadecimal Subtraction

Activity 2: Hexadecimal Calculation Subtraction :

Hex value:

$$F - B = ?$$

$$D - A = ?$$

$$E6 - 3B = ?$$

$$C2 - A9 = ?$$

$$5E - 1C = ?$$

$$D1 - 3F = ?$$

$$1A - 05 = ?$$

$$AA - 11 = ?$$

$$BF - BB = ?$$

$$C9 - 2F = ?$$

$$24 - 12 = ?$$

1.2 - Hexadecimal:

B)-Hexadecimal Subtraction

To double check your answer click in the link below:

<https://www.calculator.net/hex-calculator.html>

Hex Calculator

Hexadecimal Calculation—Add, Subtract, Multiply, or Divide

Result

Hex value:

8B – 7A = 11

Decimal value:

139 – 122 = 17

8B - 7A = ?

Calculate



Clear

1.2 - Hexadecimal:

C)-Hexadecimal Multiplication

Example 2: Hex value: $AA * A = 6A4$

Hex value: $AA = 1010\ 1010$ and Decimal value: 170

Denary	1024	512	256	128	64	32	16	8	4	2	1
Binary	0	0	0	1	0	1	0	1	0	1	0

Hex value: $A = 1010$ and Decimal value: 10

Denary	1024	512	256	128	64	32	16	8	4	2	1
Binary	0	0	0	0	0	0	0	1	0	1	0

Hex value: $AA * A = 6A4$ and Decimal value: $170 * 10 = 1700$

Denary	1024	512	256	128	64	32	16	8	4	2	1
Binary	1	1	0	1	0	1	0	0	1	0	0
	6			A			4				

1.2 - Hexadecimal:

B)-Hexadecimal Multiplication

To double check your answer click in the link below:

<https://www.calculator.net/hex-calculator.html>

Hex Calculator

Hexadecimal Calculation—Add, Subtract, Multiply, or Divide

Result

Hex value:

$AA \times A = 6A4$

Decimal value:

$170 \times 10 = 1700$

= ?

Calculate



Clear

1.2 - Hexadecimal:

C)-Hexadecimal Multiplication

Activity 3: Hexadecimal Calculation Multiplication :

Hex value:

$$2 \times A = ?$$

$$F \times F = ?$$

$$11 \times C = ?$$

$$7D \times 0A = ?$$

$$D8 \times 6 = ?$$

$$32 \times 12 = ?$$

$$B0 \times 0C = ?$$

$$EE \times 9 = ?$$

$$DA \times C = ?$$

$$FA \times B = ?$$

$$99 \times 9 = ?$$

Homework

- Complete the Sample question

● Sample question

3 a Convert the following binary numbers into hexadecimal numbers:

(i) 1 1 1 1 0 0 1 0 1 0 1 1

(ii) 1 0 1 0 1 1 1 1 1 0 1 0 1 0

b Convert the following hexadecimal numbers into binary numbers:

(i) A 1 4 A

(ii) F 9 A 8

c Convert the following hexadecimal numbers into denary numbers:

(i) 4 A 2 F

(ii) 2 D F 5

d Convert the following denary numbers into hexadecimal numbers:

(i) 6 9 4 0

(ii) 1 5 1 2 0

USES OF HEXADECIMAL



The hex number is far easier for humans to remember, copy and work with. This section reviews four uses of the hexadecimal system:

- » error codes:
- » MAC addresses
- » IPv6 addresses
- » HTML colour codes

USES OF HEXADECIMAL

- ❑ **Error codes** are often shown as hexadecimal values. These numbers refer to the memory location of the error and are usually automatically generated by the computer.
- ❑ **Media Access Control (MAC) address** refers to a number which uniquely identifies a device on a network. The MAC address refers to the network interface card (NIC) which is part of the device. The MAC address is rarely changed so that a particular device can always be identified no matter where it is. A MAC address is usually made up of 48 bits which are shown as 6 groups of two hexadecimal digits (although 64-bit addresses also exist):
- ❑ 00 – 1C – B3 – 4F – 25 – FE is the MAC address of a device produced by the Apple Corporation (code: 001CB3) with a serial number of: 4F25FE.

NN – NN – NN – DD – DD – DD

or

NN:NN:NN:DD:DD:DD

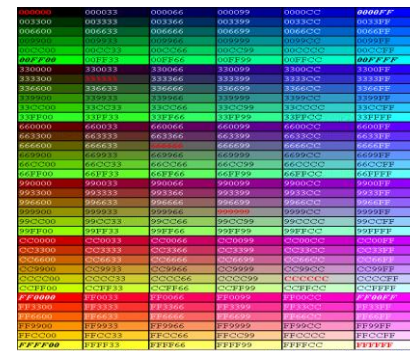
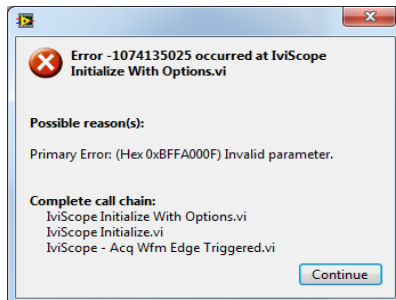
USES OF HEXADECIMAL

□ Used in error messages

- <https://www.bbc.co.uk/bitesize/guides/zp73wmn/revision/2#:~:text=Hex%20is%20often%20used%20in,find%20and%20then%20fix%20problems.>

□ Representation of colours in images and HTML

□ Challenge! – what other uses of hexadecimal in computing can you find?



USES OF HEXADECIMAL

- **Internet Protocol (IP) addresses:** Each device connected to a network is given an address known as the **Internet**
- **Protocol (IP) address.** An IPv4 address is a 32-bit number written in denary or hexadecimal form: e.g. 109.108.158.1 (or 77.76.9e.01 in hex). IPv4 has recently been improved upon by the adoption of IPv6. An IPv6 address is a 128-bit number broken down into 16-bit chunks, represented by a hexadecimal number.
- For example:

a8fb:7a88:fff0:0fff:3d21:2085:66fb:f0fa

Note IPv6 uses a colon (:) rather than a decimal point (.) as used in IPv4.

HyperText Mark-up Language (HTML)

- HYPERTEXT MARK-UP LANGUAGE (HTML) is used when writing and developing web pages. HTML isn't a programming language but is simply a mark-up language. A mark-up language is used in the processing, definition and presentation of text (for example, specifying the colour of the text).
- HTML uses `<tags>` which are used to bracket a piece of code; for example, `<td>` starts a standard cell in an HTML table, and `</td>` ends it. Whatever is between the two tags has been defined. Here is a short section of HTML code:

HyperText Mark-up Language (HTML)

□ HTML example:

```
<tr>
  <td><h3>Small car</h3>
    <h3>Used car sales</h3>
    <h2>Cars from $500</h2>
    <br><h2>Cash sales only</h2></td></br>
</tr>
<table border="1">
  <colgroup>
    <col span="2" style="background-color:red">
    <col style="background-color:yellow">
  </colgroup>
```

HyperText Mark-up Language (HTML)

- HTML code is often used to represent colours of text on the computer screen. The values change to represent different colours. The different intensity of the three primary colours (red, green and blue) is determined by its hexadecimal value.

For example:

- • # FF 00 00 represents primary colour **red**
- • # 00 FF 00 represents primary colour **green**
- • # 00 00 FF represents primary colour **blue**

HyperText Mark-up Language (HTML)

1 - Activity 1

Activity 1.8

Using the internet, find the HTML codes for a number of colours. Try entering HTML code into the computer and see how the colours and font types can be changed to good effect. Make use of websites, such as www.html.am/ to produce your own web pages.

With a little practice, you can import/embed images into your own design of web page using freely available software. Remember this is not a programming language. It is simply a mark-up language, so very little programming skill is required to use HTML.

Questions:



- Describe two applications of the hexadecimal system?
- Describe what is meant by HTML structure and presentation for a web page?

1.1 DATA REPRESENTATION

IGCSE PAST PAPER 1 EXAMS QUESTION 1:

1 a Convert the denary number 59 into 8-bit binary. [1]

--	--	--	--	--	--	--	--

b Convert the 12-bit binary number 011010010001 into denary. Show your working

..... [2]

1.1 DATA REPRESENTATION

IGCSE PAST PAPER 1 EXAMS QUESTION 2:

Jane answers an examination question about computers and data correctly. **Six** different words or numbers have been removed from her answer.

Complete the sentences in Jane's answer, using the list given. Not all items in the list need to be used.

- |
- 2
- 10
- 16
- analogue
- binary
- denary
- digital
- hexadecimal

✓
As humans, we process data, but a computer cannot process this type of data. For a computer to be able to process data it needs to be converted to data.

As humans, we mostly use a number system;

this is a base number system.

Computers use a number system;

this is a base number system.

1.1 DATA REPRESENTATION

IGCSE PAST PAPER 1 EXAMS ANSWER 2:

Jane answers an examination question about computers and data correctly.

Six different words or numbers have been removed from her answer.

Complete the sentences in Jane's answer, using the list given. Not all items in the list need to be used.

- 2
- 10
- 16
- analogue
- binary
- denary
- digital
- hexadecimal

✓

As humans, we processanalogue.....✓..... data, but a computer cannot process this type of data. For a computer to be able to process data it needs to be converted todigital...✓..... data.

As humans, we mostly use adenary...✓..... number system;

this is a base10✓..... number system.

Computers use abinary✓..... number system;

this is a base2✓..... number system.

1.1 DATA REPRESENTATION

IGCSE PAST PAPER 1 EXAMS QUESTION 3:

- (b) Pradeep has two 8-bit binary values that he needs to convert to hexadecimal values for his project.

Give the **hexadecimal** values for the two 8-bit binary values.

01010000

00111101

[4]

1.1 DATA REPRESENTATION

IGCSE PAST PAPER 1 EXAMS QUESTION 4:

(a) Give the **denary** value of each of the three 12-bit binary values.

(i) 000000001100

..... [1]

(ii) 000011000110

..... [1]

(iii) 010011000001

..... [1]

Working space

.....

.....

.....

.....

.....

.....

.....

1.1 DATA REPRESENTATION

IGCSE PAST PAPER 1 EXAMS QUESTION 4:

A digital alarm clock is controlled by a microprocessor. It uses the 24-hour clock system (i.e. 6 pm is 18:00).

Each digit in a typical display is represented by a 4-digit binary code.

For example:



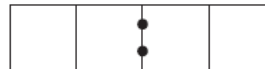
(clock display)

is represented by:

0	0	0	0	1st digit (0)
1	0	0	0	2nd digit (8)
0	0	1	1	3rd digit (3)
0	1	0	1	4th digit (5)

(a) What time is shown on the clock display if the 4-digit binary codes are:

0	0	0	1
0	1	1	0
0	1	0	0
1	0	0	1



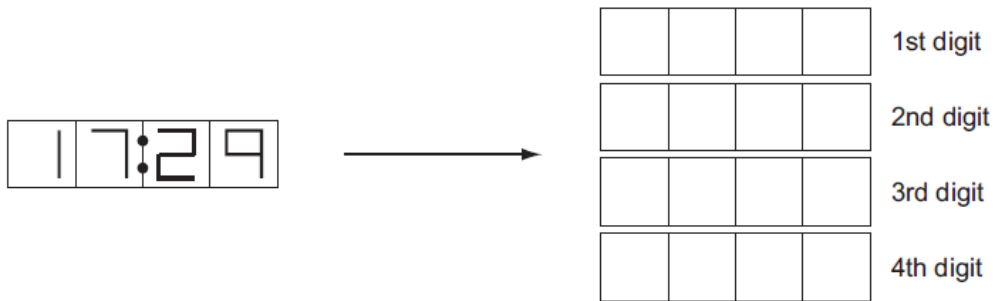
(clock display)

[2]

1.1 DATA REPRESENTATION

IGCSE PAST PAPER 1 EXAMS QUESTION 4:

(b) What would be stored in the 4-digit binary codes if the clock display time was:



[4]

1.1 DATA REPRESENTATION

IGCSE PAST PAPER 1 EXAMS QUESTION 5:

Convert the following hexadecimal number into denary

5F

Convert the following binary number into hexadecimal

101011000100

Convert the following hexadecimal number into binary

3ED

Number systems: One's and Two's Compliment

Up until now we have assumed all binary numbers have positive values. There are a number of methods to represent both positive and negative numbers. We will consider:

- Signed Magnitude method
- one's complement
- two's complement.

It is not possible to add minus or plus symbol in front of a binary number because a binary number can have only two symbol either 0 or 1 for each position or bit. That's why we use this extra bit called sign bit or sign flag. The value of sign bit is 1 for negative binary numbers and 0 for positive numbers.

When an integer binary number is positive, the sign is represented by 0 and the magnitude by a positive binary number. When the number is negative, the sign is represented by 1 but the rest of the number may be represented in one of three possible ways: Sign-Magnitude method, 1's Complement method, and 2's complement method. These are explained as following below.

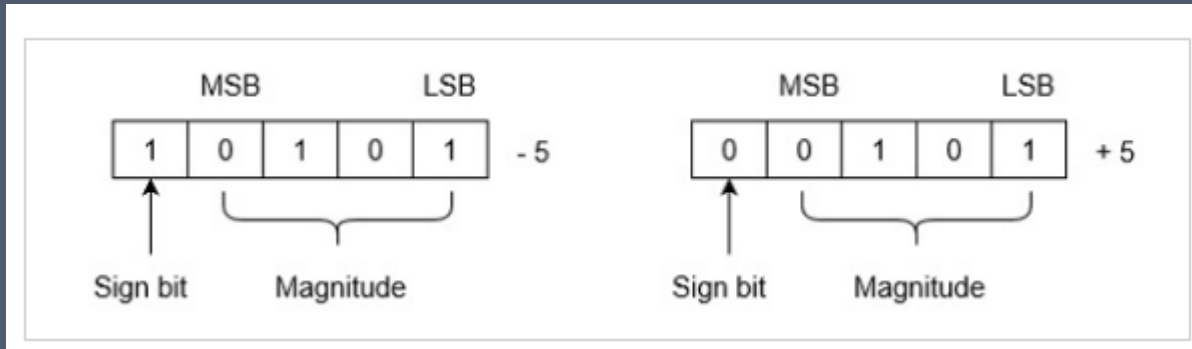
Number systems: One's and Two's Complement

a). Signed Magnitude method

In this method, number is divided into two parts: Sign bit and Magnitude. If the number is positive then sign bit will be 0 and if number is negative then sign bit will be 1. Magnitude is represented with the binary form of the number to be represented.

Example 1: Let we are using **5 bits register**. The representation of -5 to +5 will be as follows:

Most significant bit (MSB) and Least significant bit (LSB)



Number systems: One's and Two's Complement

a). Signed Magnitude method

Example 1: Let we are using **4 binary bits register**. The representation of -7 to +7 will be as follows:

Signed Magnitude

(-6)

2^{n-1}
 $2^{4-1} = 2^3 = 8$

6 →

8	4	2	1
0	1	1	0

↑ ↑
Sign M

-6 1 1 1 0

0 → +
1 → -

1.1 Data Representation

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Number systems: One's and Two's Complement

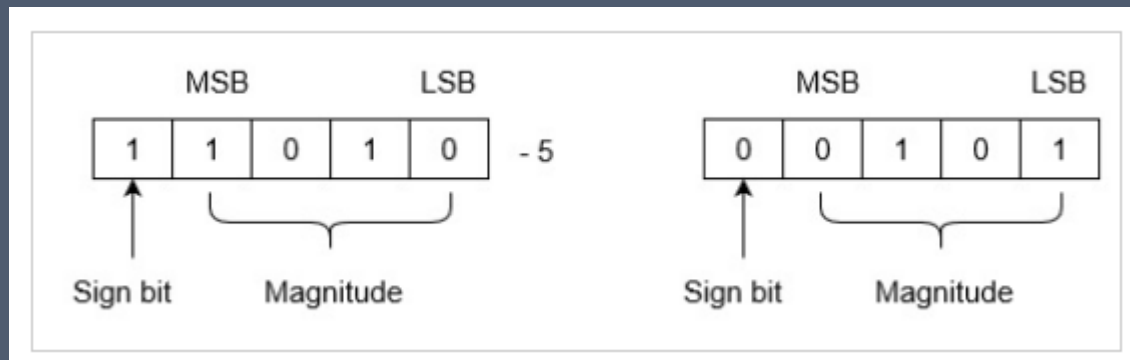
b). One's Complement Method:

Positive numbers are represented in the same way as they are represented in sign magnitude method. If the number is negative then it is represented using 1's complement. First represent the number with positive sign and then take 1's complement of that number.

In **one's complement**, each digit in the binary number is inverted (in other words, 0 becomes 1 and 1 becomes 0).

For example 1: 01000110 (denary value 70) becomes 10111001 (denary value -70).

For example 2: Let we are using 5 bits register. The representation of -5 and +5 will be as follows



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c). Two's Complement Method:

2's Complement

	-8	4	2	1	
7	0	1	1	1	
	1	0	0	0	← 1's
					← 2's
	+ _____ +				
-7		1	0	0	1
		-8 + 0 + 0 + 1 = -7			

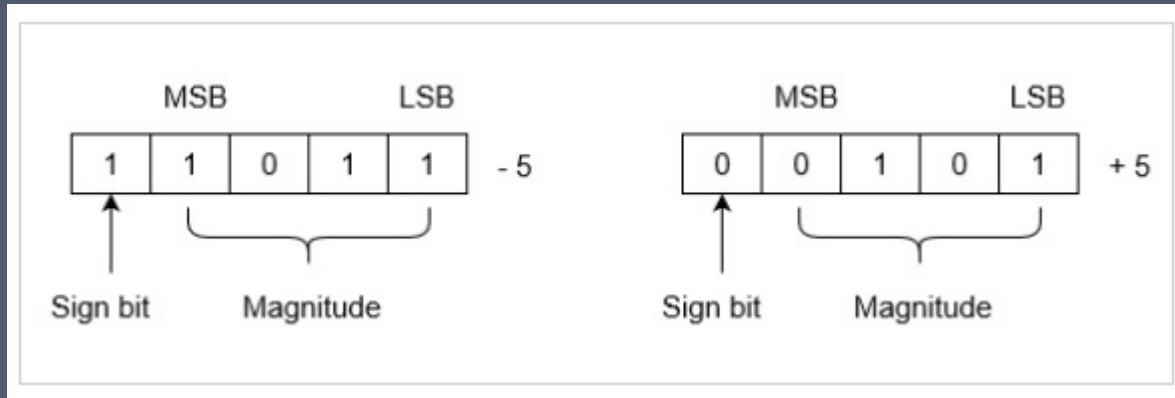
1.1 Data Representation

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Number systems: One's and Two's Complement

c). Two's Complement Method:

Example 2: Let we are using 5 bits registers. The representation of -5 and +5 will be as follows:



+5 is represented as it is represented in sign magnitude method. -5 is represented using the following steps:

(i) $+5 = 0\ 0101$

(ii) Take 2's complement of $0\ 0101$ and that is $1\ 1011$. MSB is 1 which indicates that number is negative.

MSB is always 1 in case of negative numbers

1.1 Data Representation

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Number systems: One's and Two's Complement

c). Two's Complement Method:

Example 2: Let we are using 8 bits registers. The representation of -104 and +104 will be as follows:

+104		128	64	32	16	8	4	2	1
Binary		0	1	1	0	1	0	0	0
1 comp Opposite		1	0	0	1	0	1	1	1
Add 1		0	0	0	0	0	0	0	1
- 104		1	0	0	1	0	0	0	0

TWO'S COMPLEMENT

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$\boxed{-6}$

$(128 - 6 = 122)$

-128	64	32	16	8	4	2	1
1	1	1	1	1	0	1	0

$-128 + 64 + 32 + 16 + 8 + 2 = -6$

$\boxed{-49}$

$(128 - 49 = 79)$

-128	64	32	16	8	4	2	1
1	1	0	0	1	1	1	1

$-128 + 64 + 8 + 4 + 2 + 1 = -49$

$\boxed{-109}$

$(128 - 109 = 19)$

-128	64	32	16	8	4	2	1
1	0	0	1	0	0	1	1

$-128 + 16 + 2 + 1 = -109$

$\boxed{-77}$

$(128 - 77 = 51)$

-128	64	32	16	8	4	2	1
1	0	1	1	0	0	1	1

$-128 + 32 + 16 + 2 + 1 = -77$

$\boxed{-17}$

$(128 - 17 = 111)$

-128	64	32	16	8	4	2	1
1	1	1	0	1	1	1	1

$-128 + 64 + 32 + 8 + 4 + 2 + 1 = -17$

$\boxed{-122}$

$(128 - 122 = 6)$

-128	64	32	16	8	4	2	1
1	0	0	0	0	1	1	0

$-128 + 4 + 2 = -122$

TWO'S COMPLEMENT

WRITING NEGATIVE BINARY NUMBERS IN TWO'S COMPLEMENT FORMAT AND CONVERTING TO DENARY



Example 1

The following three examples show how we can write negative binary numbers in the two's complement format:

-128	64	32	16	8	4	2	1
1	0	0	1	0	0	1	1

By following our normal rules, each time a 1 appears in a column, the column value is added to the total. So, we can see that in denary this is: $-128 + 16 + 2 + 1 = -109$.

-128	64	32	16	8	4	2	1
1	1	1	0	0	1	0	0

Similarly, in denary this number is $-128 + 64 + 32 + 4 = -28$.

-128	64	32	16	8	4	2	1
1	1	1	1	0	1	0	1

This number is equivalent to $-128 + 64 + 32 + 16 + 4 + 1 = -11$.

TWO'S COMPLEMENT

WRITING NEGATIVE BINARY NUMBERS IN TWO'S COMPLEMENT FORMAT AND CONVERTING TO DENARY

Consider the number +67 in 8-bit (two's complement) binary format:

-128	64	32	16	8	4	2	1
0	1	0	0	0	0	1	1

Method 1

Now let's consider the number -67. One method of finding the binary equivalent to -67 is to simply put 1s in their correct places:

-128	64	32	16	8	4	2	1
1	0	1	1	1	1	0	1

$$-128 + 32 + 16 + 8 + 4 + 1 = -67$$

Method 2

However, looking at the two binary numbers above, there is another possible way to find the binary representation of a negative denary number:

first write the number as a positive binary value – in this case 67: 0 1 0 0 0 0 1 1

we then invert each binary value, which means swap the 1s and 0s around: 1 0 1 1 1 1 0 0

then add 1 to that number: 1

this gives us the binary for -67: 1 0 1 1 1 1 0 1

TWO'S COMPLEMENT

WRITING NEGATIVE BINARY NUMBERS IN TWO'S COMPLEMENT FORMAT AND CONVERTING TO DENARY



Example 2

Convert -79 into an 8-bit binary number using two's complement format.

Method 1

As it is a negative number, we need a 1-value in the -128 column.

-79 is the same as $-128 + 49$

We can make up 49 from $32 + 16 + 1$; giving:

-128	64	32	16	8	4	2	1
1	0	1	1	0	0	0	1

Method 2

write 79 in binary:	0 1 0 0 1 1 1 1
invert the binary digits:	1 0 1 1 0 0 0 0
add 1 to the inverted number	1
thus giving -79 :	1 0 1 1 0 0 0 1

-128	64	32	16	8	4	2	1
1	0	1	1	0	0	0	1

It is a good idea to practise both methods.

TWO'S COMPLEMENT

WRITING NEGATIVE BINARY NUMBERS IN TWO'S COMPLEMENT FORMAT AND CONVERTING TO DENARY

? Example 3

The following 4-bit binary number represents denary number 6:

-8	4	2	1
0	1	1	0

Applying two's complement ($1\ 0\ 0\ 1 + 1$) would give:

-8	4	2	1
1	0	1	0

in other words: -6

TWO'S COMPLEMENT

WRITING NEGATIVE BINARY NUMBERS IN TWO'S COMPLEMENT FORMAT AND CONVERTING TO DENARY

Activity 1.13

Convert the following negative denary numbers into binary numbers using the two's complement format:

a	-18	c	-47	e	-88	g	-100	i	-16
b	-31	d	-63	f	-92	h	-1	j	-127

Activity 1.14

Convert the following negative binary numbers (written in two's complement format) into negative denary numbers:

a	1	1	0	0	1	1	0	1
b	1	0	1	1	1	1	1	0
c	1	1	1	0	1	1	1	1
d	1	0	0	0	0	1	1	1
e	1	0	1	0	0	0	0	0
f	1	1	1	1	1	0	0	1
g	1	0	1	0	1	1	1	1
h	1	1	1	1	1	1	1	1
i	1	0	0	0	0	0	0	1
j	1	1	1	1	0	1	1	0

Number systems: One's and Two's Complement:

Watch the video for the Signed Binary (Sign and Magnitude & Two's Complement)

<https://www.youtube.com/watch?v=sJXTo3EZoxM>

Binary - 2's Complement

Signed Magnitude:

$+7 \rightarrow 00111$
 $-7 \rightarrow \boxed{1}0111$

2's Complement:

$+12 \rightarrow 01100$
 $-12 \rightarrow 10100$

0100111
 -0011001

0100111
 $+1100111$

1.1 DATA REPRESENTATION

WATCH THE VIDEO FOR THE TWO'S COMPLEMENT REPRESENTATION OF A SIGNED INTEGER

[HTTPS://WWW.YOUTUBE.COM/WATCH?V=MRVCGIJXI9W](https://www.youtube.com/watch?v=MRVCGIJXI9W)

Convert -6 into 8 bit binary

-128	64	32	16	8	4	2	1
1	1	1	1	1	0	1	0

$$-128 + 64 + 32 + 16 + 8 + 2 = -6$$

$$-6_{10} = 11111010_2$$

1.1 DATA REPRESENTATION

NUMBER SYSTEMS: ONE'S AND TWO'S COMPLIMENT:

Convert the following negative Decimal number to binary

$(-49) = ?$

$(-109) = ?$

$(-107) = ?$

$(-77) = ?$

$(-17) = ?$

$(-122) = ?$

$(-38) = ?$

$(-200) = ?$

$(-26) = ?$

$(-2) = ?$

$(-11) = ?$

$(-52) = ?$

COMPUTER SCIENCE

CHAPTER 1:

1.1 DATA REPRESENTATION

1.2 TEXT, SOUND AND IMAGES

By: Nouredine Tadjerout

Learning Objectives

1.2 Text, sound and images

- 1 Understand how and why a computer represents text and the use of character sets, including American standard code for information interchange (ASCII) and Unicode
 - Text is converted to binary to be processed by a computer
 - Unicode allows for a greater range of characters and symbols than ASCII, including different languages and emojis
 - Unicode requires more bits per character than ASCII
- 2 Understand how and why a computer represents sound, including the effects of the sample rate and sample resolution
 - A sound wave is sampled for sound to be converted to binary, which is processed by a computer
 - The sample rate is the number of samples taken in a second
 - The sample resolution is the number of bits per sample
 - The accuracy of the recording and the file size increases as the sample rate and resolution increase
- 3 Understand how and why a computer represents an image, including the effects of the resolution and colour depth
 - An image is a series of pixels that are converted to binary, which is processed by a computer
 - The resolution is the number of pixels in the image
 - The colour depth is the number of bits used to represent each colour
 - The file size and quality of the image increases as the resolution and colour depth increase

ASCII CODE

<https://www.youtube.com/watch?v=H4l42nbYmrU>

ASCII Code

Char.	ASCII	Char.	ASCII	Char.	ASCII
@	64	U	85	j	106
A	65	V	86	k	107
B	66	W	87	l	108
C	67	X	88	m	109
D	68	Y	89	n	110
E	69	Z	90	o	111
F	70	[	91	p	112
G	71	\	92	q	113
H	72	]	93	r	114
I	73	^	94	s	115
J	74	_	95	t	116
K	75	`	96	u	117
L	76	a	97	v	118
M	77	b	98	w	119
N	78	c	99	x	120
O	79	d	100	y	121
P	80	e	101	z	122
Q	81	f	102	[	123
R	82	g	103	\	124
S	83	h	104	]	125
T	84	i	105	^	126

B → 1 0 0 0 0 1 0

L → 1 1 0 1 1 0 0

U → 1 1 1 0 1 0 1

e → 1 1 0 0 1 0 1

ASCII CODE Table :

Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
32	20	<SPACE>	64	40	@	96	60	`
33	21	!	65	41	A	97	61	a
34	22	"	66	42	B	98	62	b
35	23	#	67	43	C	99	63	c
36	24	\$	68	44	D	100	64	d
37	25	%	69	45	E	101	65	e
38	26	&	70	46	F	102	66	f
39	27	'	71	47	G	103	67	g
40	28	(	72	48	H	104	68	h
41	29	)	73	49	I	105	69	i
42	2A	*	74	4A	J	106	6A	j
43	2B	+	75	4B	K	107	6B	k
44	2C	,	76	4C	L	108	6C	l
45	2D	-	77	4D	M	109	6D	m
46	2E	.	78	4E	N	110	6E	n
47	2F	/	79	4F	O	111	6F	o
48	30	0	80	50	P	112	70	p
49	31	1	81	51	Q	113	71	q
50	32	2	82	52	R	114	72	r
51	33	3	83	53	S	115	73	s
52	34	4	84	54	T	116	74	t
53	35	5	85	55	U	117	75	u
54	36	6	86	56	V	118	76	v
55	37	7	87	57	W	119	77	w
56	38	8	88	58	X	120	78	x
57	39	9	89	59	Y	121	79	y
58	3A	:	90	5A	Z	122	7A	z
59	3B	;	91	5B	[	123	7B	{
60	3C	<	92	5C	\	124	7C	
61	3D	=	93	5D	]	125	7D	}
62	3E	>	94	5E	^	126	7E	~
63	3F	?	95	5F	_	127	7F	<DELETE>

ASCII CODE Table :

ASCII Code: Character to Binary

0	0011 0000	O	0100 1111	m	0110 1101
1	0011 0001	P	0101 0000	n	0110 1110
2	0011 0010	Q	0101 0001	o	0110 1111
3	0011 0011	R	0101 0010	p	0111 0000
4	0011 0100	S	0101 0011	q	0111 0001
5	0011 0101	T	0101 0100	r	0111 0010
6	0011 0110	U	0101 0101	s	0111 0011
7	0011 0111	V	0101 0110	t	0111 0100
8	0011 1000	W	0101 0111	u	0111 0101
9	0011 1001	X	0101 1000	v	0111 0110
A	0100 0001	Y	0101 1001	w	0111 0111
B	0100 0010	Z	0101 1010	x	0111 1000
C	0100 0011	a	0110 0001	y	0111 1001
D	0100 0100	b	0110 0010	z	0111 1010
E	0100 0101	c	0110 0011	.	0010 1110
F	0100 0110	d	0110 0100	,	0010 0111
G	0100 0111	e	0110 0101	:	0011 1010
H	0100 1000	f	0110 0110	;	0011 1011
I	0100 1001	g	0110 0111	?	0011 1111
J	0100 1010	h	0110 1000	!	0010 0001
K	0100 1011	I	0110 1001	'	0010 1100
L	0100 1100	j	0110 1010	"	0010 0010
M	0100 1101	k	0110 1011	{	0010 1000
N	0100 1110	l	0110 1100	}	0010 1001
				space	0010 0000

INTERNAL CODING OF NUMBERS

WATCH THE VIDEO FOR THE REPRESENTING TEXT IN BINARY (ASCII & UNICODE)

[HTTPS://WWW.YOUTUBE.COM/WATCH?V=TDmEXCDX-UC](https://www.youtube.com/watch?v=TDmEXCDX-UC)

ASCII

- Characters are encoded with the use of **character sets**. These are a group of codes that assign a character to a unique bit pattern e.g.:

- ASCII is a character set developed in 1960 in an attempt to solve compatibility issues

- Originally** its character codes were 7 bits long, but then it was **extended** to have a bit length of 8. **Original = 128 characters, extended = 256.**

WATCH THE VIDEO FOR THE ASCII AND UNICODE CHARACTER SETS¹⁵²

[HTTPS://WWW.YOUTUBE.COM/WATCH?V=5AJKKGSEUNY](https://www.youtube.com/watch?v=5AJKKGSEUNY)



32 BITS = 2,147,483,647



ASCII AND UNICODE CHARACTER SETS

ASCII

THE ASCII CHARACTER SET IS A **7-BIT SET** OF CODES THAT ALLOWS **128 DIFFERENT CHARACTERS**. THAT IS ENOUGH FOR EVERY UPPER-CASE LETTER, LOWER-CASE LETTER, DIGIT AND PUNCTUATION MARK ON MOST KEYBOARDS. ASCII IS ONLY USED FOR THE ENGLISH LANGUAGE.

THIS TABLE SHOWS SOME EXAMPLES OF LETTERS REPRESENTED USING THE ASCII CHARACTER SET:

Character	Denary value	Binary value	Hex
N	78	1001110	4E
O	79	1001111	4F
P	80	1010000	50
Q	81	1010001	51
R	82	1010010	52

EXTENDED ASCII

EXTENDED ASCII CODE IS AN 8-BIT CHARACTER SET THAT REPRESENTS **256 DIFFERENT CHARACTERS**, MAKING IT POSSIBLE TO USE CHARACTERS SUCH AS É OR ©. EXTENDED ASCII IS USEFUL FOR EUROPEAN LANGUAGES

ASCII AND UNICODE CHARACTER SETS

CONSIDER THE UPPERCASE AND LOWERCASE CODES IN BINARY OF CHARACTERS. FOR EXAMPLE

'a'	1	1	0	0	0	0	1	hex 61 (lower case)
'A'	1	0	0	0	0	0	1	hex 41 (upper case)
'y'	1	1	1	1	0	0	1	hex 79 (lower case)
'Y'	1	0	1	1	0	0	1	hex 59 (upper case)

ASCII code has a number of disadvantages. The main disadvantage is that it does not represent characters in non-Western languages, for example Chinese characters. For this reason, different methods of coding have been developed over the years. One coding system is called **Unicode**

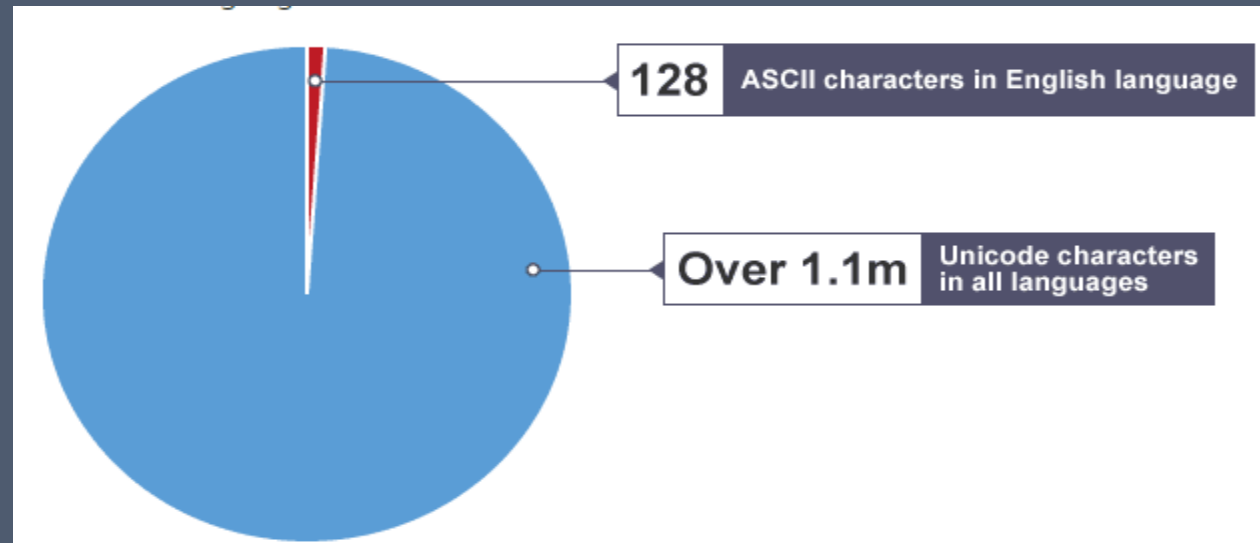
Unicode can represent all languages of the world, thus supporting many operating systems, search engines and internet browsers used globally.

ASCII AND UNICODE CHARACTER SETS

UNICODE

UNICODE USES BETWEEN **8** AND **32 BITS PER CHARACTER**, SO IT CAN REPRESENT CHARACTERS FROM LANGUAGES FROM ALL AROUND THE WORLD. IT IS COMMONLY USED ACROSS THE INTERNET. AS IT IS LARGER THAN ASCII, IT MIGHT TAKE UP MORE **STORAGE** SPACE WHEN SAVING DOCUMENTS.

GLOBAL COMPANIES, LIKE FACEBOOK AND GOOGLE, WOULD NOT USE THE ASCII CHARACTER SET BECAUSE THEIR USERS COMMUNICATE IN MANY DIFFERENT LANGUAGES.



1.2 Text, sound and images:

size Files

Text and numbers can be stored in a number of formats. Text is usually stored in an ASCII format. If ASCII format is used to store text on a file, then the ASCII table is used to store each of the characters. For example, the word **COMPUTER** would be stored as either: **67 79 77 80 85 84 69 82** or **%43 %4F %4D %50 %55 %54 %45 %52**

(the first code is in denary and the second in hexadecimal).

Text files can also undergo file compression. These use complex algorithms that work on redundancy or repeated sections of words (e.g. OU in yOUr, cOUtry or mOUntain). The following section shows, in very simple terms, how this could work: The phrase '**THIS SECTION SHOWS YOU HOW THIS WOULD WORK**' consists of 35 memory units (ignoring spaces). Repeated words, such as 'THIS' could be put into a data dictionary and be replaced by '1'. Repeated word sections, such as 'HOW' and 'OU' could be replaced by the numbers '2' and '3'. Our phrase then becomes '**1 SECTION S2S Y3 2 1 W3LD WORK**'. Again, ignoring spaces, this compressed form now uses only 23 memory units. This is about a 33% saving in file size. Obviously if whole pages are to be stored, then repeated words and word sections become even more numerous. Compression algorithms take many things into account when creating these compressed files – but this is outside the scope of this textbook. This is clearly an example of lossless compression since the original phrase can be reformed if necessary using the data dictionary and compressed file.

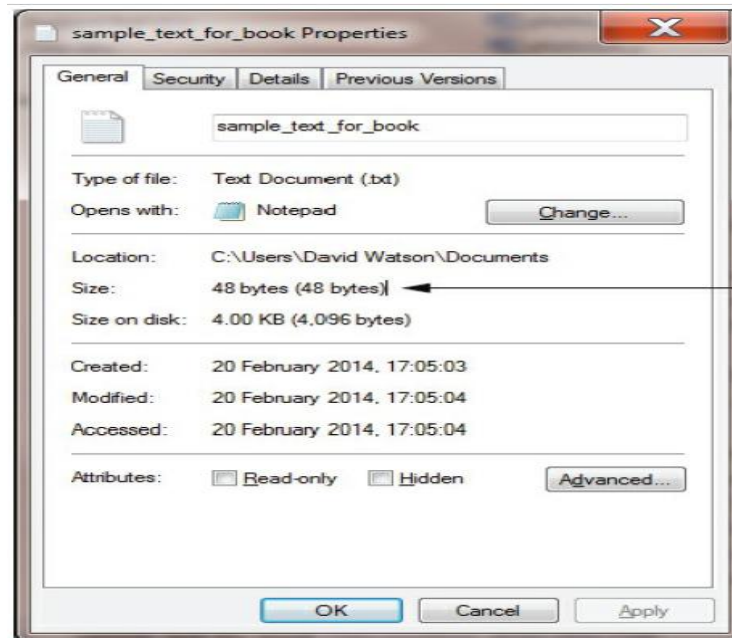
Noureddine Tadjerout

1.2 Text, sound and images:

How to estimate the size of a file

Estimating the size of a text file is relatively straightforward. In [Chapter 1](#) we introduced the idea of ASCII code. Each character from the keyboard has a value of 1 byte. Suppose we typed in the following message:

This is text from the computer science text book = 48 words = 48 bytes and then saved it under the filename 'sample_text_for_book'. If we then looked at the properties of the file just saved, we would see:



This value shows the size of the file just stored

1.2 Text, sound and images:

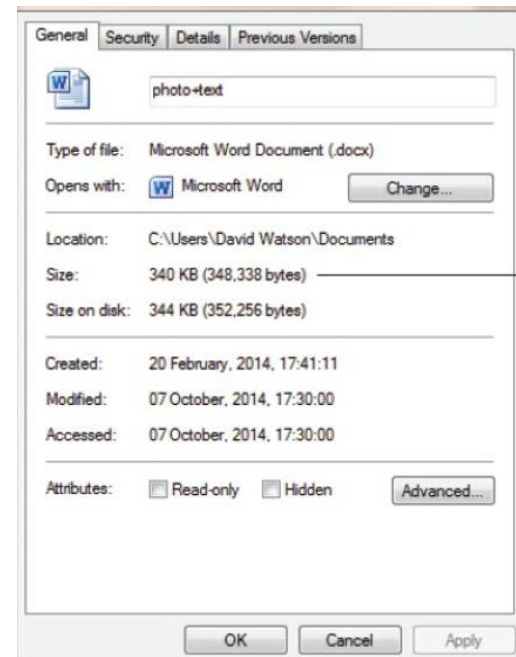
If we count the number of characters in the text typed in, we get the number 48. This exactly matches the file size shown in the screenshot.

Earlier on in the chapter we looked at file sizes for photos and for sound. This is obviously a little more complex, but we saw that one pixel, for example, occupies **3 bytes of memory**. Suppose we imported the photograph shown below:



Looking at the properties of the photograph, we find this:

The photograph was 424 by 256 pixels which confirms the file size of 325 632 bytes (318 kilobytes).

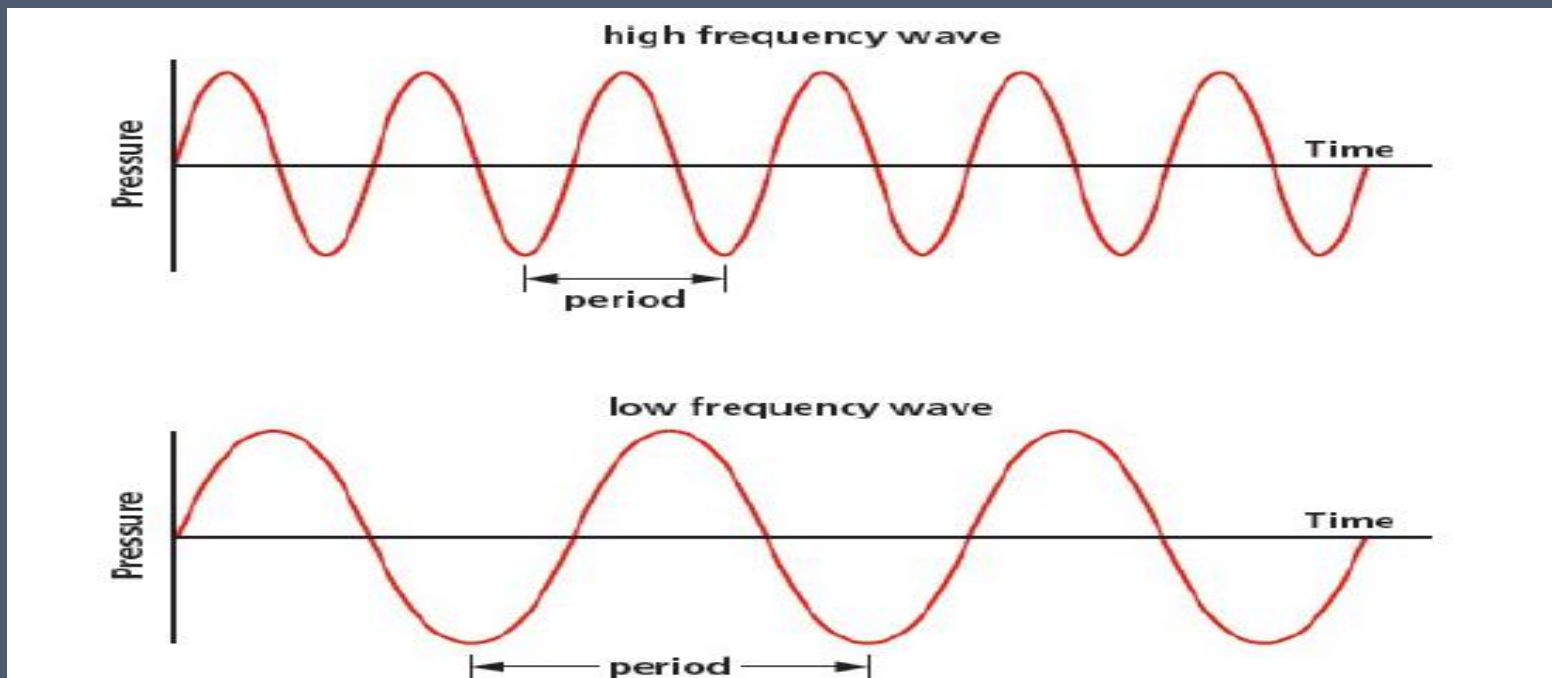


File size is 318 kilobytes for the actual picture and 22 kilobytes for the codes used

1.2 Text, sound and images:

Sound :

Sound requires a medium in which to travel through (it cannot travel in a vacuum). This is because it is transmitted by causing oscillations of particles within the medium. The human ear picks up these oscillations (changes in air pressure) and interprets them as sound. Each sound wave has a frequency and wavelength; the amplitude specifies the loudness of the sound.



WATCH THE VIDEO FOR THE REPRESENTING SOUND IN BINARY

[HTTPS://WWW.YOUTUBE.COM/WATCH?V=HLOTUCFTUV8](https://www.youtube.com/watch?v=HLOTUCFTUV8) (4 MINUTES)

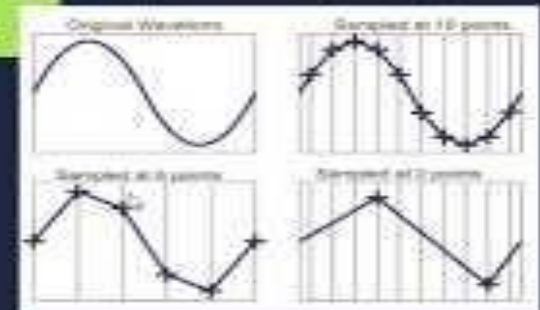
Digitising Sound **Sampling = recording a signal at regular intervals**

- Generally done by taking a sample of the amplitude of the sound wave, at regular time intervals. A binary value is given to each reading.



The sampling frequency is the number of samples obtained per second. It is measured in Hertz (Hz).

- The greater the sampling frequency, the truer the digital signal is to the original, but the greater the file size.



1.2 TEXT, SOUND AND IMAGES:

SOUND :

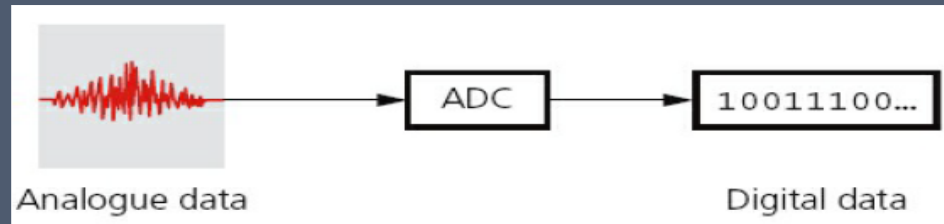
- SOUND ARISES FROM PRESSURE CHANGES
- SOUND IS AN IRREGULAR ANALOGUE WAVEFORM
- SOUND HAS TO BE SAMPLED TO BE STORED
- SAMPLING HAS A DEFINED SAMPLING RESOLUTION AND A DEFINED SAMPLING RATE
- A SOUND ENCODER CONTAINS A BAND-LIMITING FILTER AND AN ANALOGUE TO DIGITAL CONVERTER

1.2 Text, sound and images:

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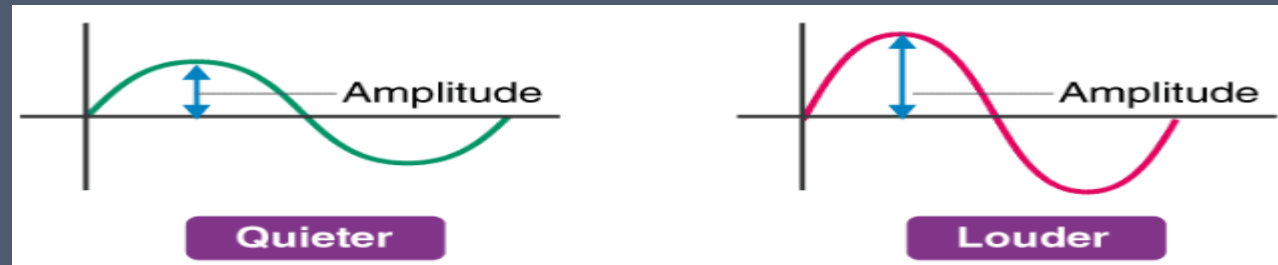
Sound :

Sound is an analogue value; this needs to be digitized in order to store sound in a computer. This is done using an **analogue to digital converter (ADC)**. If the sound is to be used as a music file, it is often filtered first to remove higher frequencies and lower frequencies which are outside the range of human hearing. To convert the analogue data to digital, the sound waves are sampled at a given time rate. The amplitude of the sound cannot be measured precisely, so approximate values are stored.



What is Amplitude of Sound?

The amplitude of a sound wave is the measure of the **height** of the **wave**. The amplitude of a sound wave can be defined as the loudness (Volume) or the amount of maximum displacement of vibrating particles of the medium from their mean position when the sound is produced. It is the distance between the crest or trough and the mean position of the wave and its measure in decibel (**db**),



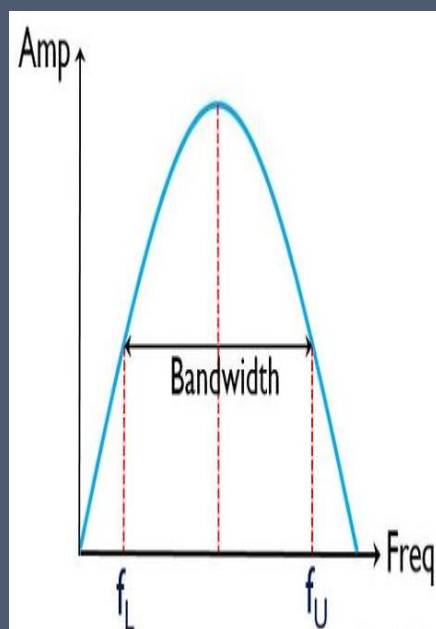
1.2 Text, sound and images:

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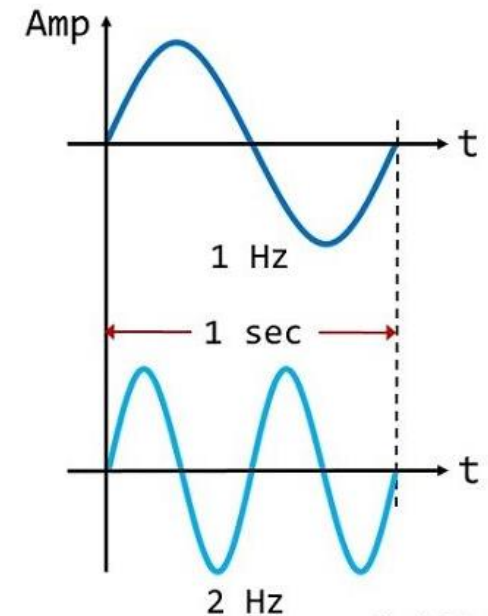
Sound :

What is bandwidth of frequency?

Bandwidth is **a range of frequencies within a continuous set of frequencies**. It is measured in Hertz (Hz). The purpose of a communication system is to transfer information from the transmitter which is located in one place to a receiver which is usually far away from the transmitter.



Basis for Comparison	Frequency	Bandwidth
Basic	It defines the number of complete cycles in unit time.	It defines the amount of data transmitted in unit time.
Specified in terms of	cycles/sec	bits/sec
Denoted as	f	B



1.2 Text, sound and images:

Sound :

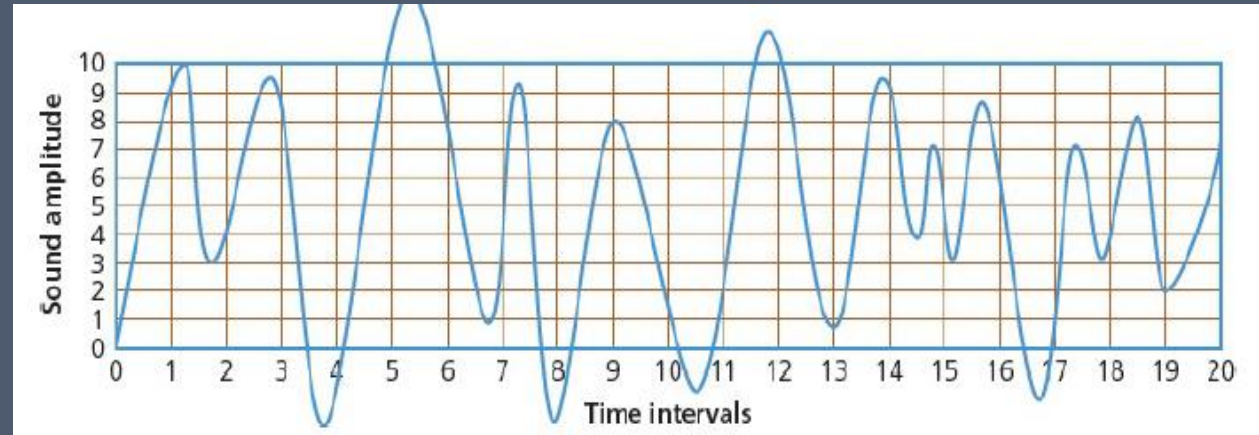


Figure above shows a sound wave. The x-axis shows the time intervals when the sound was sampled (**0** to **20**), and the y-axis shows the amplitude of the sampled sound (the amplitudes above **10** and below **0** are filtered out in this example).

At time interval **1**, the approximate amplitude is **9**; at time interval **2**, the approximate amplitude is **4**, and so on for all **20** time intervals. Because the amplitude range in Figure above is **0** to **10**, then **4** binary bits can be used to represent each amplitude value (for example, **9** would be represented by the binary value **1001**).

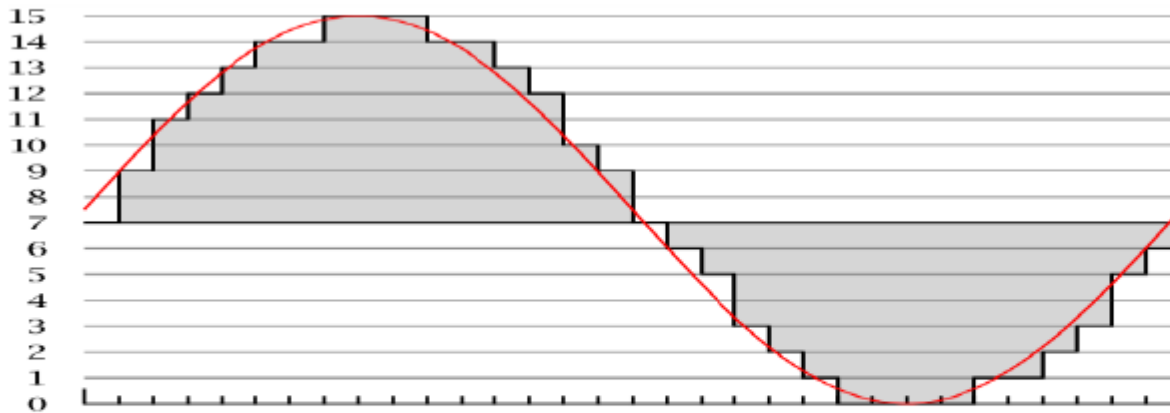
Increasing the number of possible values used to represent sound amplitude also increases the accuracy of the sampled sound (for example, using a range of **0** to **127** gives a much more accurate representation of the sound sample than using a range of, for example, **0** to **10**). This is known as the **sampling resolution** (also known as the **bit depth**).

1.2 Text, sound and images:

Sound :

The process of **converting an analogue signal** into a **digital** one is known as **sampling**.

Using this image we can count that there have been **32 samples** taken



We can also see, using the y axis that there are **16 possible sound states** which needs to be represented using **4 bits per sample**.

To calculate the size of the sound file you simply need to **multiply** the bit depth by the number of samples..

$$32 \times 4 = 128 \text{ bits}$$

$$128 / 8 = 16 \text{ Bytes}$$

1.2 Text, sound and images:

Sound :

The benefits and drawbacks of using a larger sampling resolution when recording sound

Benefits	Drawbacks
larger dynamic range	produces larger file size
better sound quality	takes longer to transmit/download music files
less sound distortion	requires greater processing power

1.2 Text, sound and images:

Sound :

Sampling rate is the number of sound samples taken per second. The higher the sampling rate and/or sampling resolution, the greater the file size. For example, a 16-bit sampling resolution is used when recording CDs to give better sound quality.

Let's explore more about sampling by clicking the link below:

<https://www.bbc.co.uk/bitesize/guides/z7vc7ty/revision/2>

Images:

BIT-MAP IMAGES ARE MADE UP OF PIXELS (PICTURE ELEMENTS); THE IMAGE IS STORED IN A TWO DIMENSIONAL MATRIX OF PIXELS.

When storing images as pixels, we have to consider

- at least 8 bits (1 byte) per pixel are needed to code a coloured image (this gives 256 possible colours by varying the intensity of the blue, green and red elements)
- true colour requires 3 bytes per pixel (24 bits), which gives more than one million colours
- the number of bits used to represent a pixel is called the **colour depth**.

We will now consider the actual image itself and how it can be displayed on a screen. There are two important definitions here:

- **Image resolution** refers to the number of pixels that make up an image; for example, an image could contain 4096×3192 pixels (12 738 656 pixels in total).
- **Screen resolution** refers to the number of horizontal pixels and the number of vertical pixels that make up a screen display (for example, if the screen resolution is smaller than the image resolution then the whole image cannot be shown on the screen or the original image will now be a lower quality).

WATCH THE VIDEO FOR THE BITMAP IMAGES (6 MINUTES)

[HTTPS://WWW.YOUTUBE.COM/WATCH?V=0KMIMFOALTU](https://www.youtube.com/watch?v=0KMIMFOALTU)

Bitmaps in Colour

24 bits used to describe the colour of each pixel

$2^{24} = 16,777,216$ colours



$960 * 640 = 614,400$ pixels

Each pixel requires 3 bytes

$614,400 * 3 = 1,843,200$ bytes

$1,843,200 / 1024 = 1,800$ KB

$1800 / 1024 = 1.76$ MB

WATCH THE VIDEO FOR THE REPRESENTING IMAGES IN BINARY (5MINUTES)

[HTTPS://WWW.YOUTUBE.COM/WATCH?V=A3Y_ZVOR0K0](https://www.youtube.com/watch?v=A3Y_ZVOR0K0)



Colour Depth: 4

of colours: $2^4 = 16$



8

$2^8 = 256$



24

$2^{24} = \sim 17 \text{ mil}$

IEC (INTERNATIONAL ELECTROTECHNICAL COMMISSION)

Name of memory size	Number of bytes	Equivalent denary value
1 kibibyte (1 KiB)	2^{10}	1 024 bytes
1 mebibyte (1 MiB)	2^{20}	1 048 576 bytes
1 gibibyte (1 GiB)	2^{30}	1 073 741 824 bytes
1 tebibyte (1 TiB)	2^{40}	1 099 511 627 776 bytes
1 pebibyte (1 PiB)	2^{50}	1 125 899 906 842 624 bytes
1 exbibyte (1 EiB)	2^{60}	1 152 921 504 606 846 976 bytes

This system is more accurate. Internal memories (such as RAM and ROM) should be measured using the IEC system. A 64 GiB RAM could, therefore, store 64×2^{30} bytes of data (68 719 476 736 bytes).

CALCULATING BIT-MAP IMAGE FILE SIZES

IT IS POSSIBLE TO ESTIMATE THE FILE SIZE NEEDED TO STORE A BIT-MAP IMAGE. THE FILE SIZE WILL NEED TO TAKE INTO ACCOUNT THE IMAGE RESOLUTION AND BIT DEPTH.

The file size of an image is calculated as:

image resolution (in pixels) $\times$ colour depth (in bits)

The size of a mono sound file is calculated as:

sample rate (in Hz) $\times$ sample resolution (in bits) $\times$ length of sample (in seconds)

For a stereo sound file, you would then multiply the result by two.

EXAMPLE 1: A FULL SCREEN WITH A RESOLUTION OF 1920×1080 PIXELS AND A BIT DEPTH OF 24 REQUIRES $1920 \times 1080 \times 24$ BITS = 49 766 400 BITS FOR THE FULL SCREEN IMAGE

DIVIDING BY 8 GIVES US 6 220 800 BYTES (EQUIVALENT TO 6.222 MB USING THE SI UNITS OR 5.933 MIB USING IEE UNITS). AN IMAGE WHICH DOES NOT OCCUPY THE FULL SCREEN WILL OBVIOUSLY RESULT IN A SMALLER FILE SIZE.

NOTE: WHEN SAVING A BIT-MAP IMAGE, IT IS IMPORTANT TO INCLUDE A FILE HEADER; THIS WILL CONTAIN ITEMS SUCH AS FILE TYPE (.BMP OR .JPEG), FILE SIZE, IMAGE RESOLUTION, BIT DEPTH (USUALLY 1, 8, 16, 24 OR 32), ANY TYPE OF DATA COMPRESSION EMPLOYED AND SO ON.

CALCULATING BIT-MAP IMAGE FILE SIZES

EXAMPLE 2:

A photograph is 1024×1080 pixels and uses a colour depth of 32 bits. How many photographs of this size would fit onto a memory stick of 64 GiB?

- 1 Multiply number of pixels in vertical and horizontal directions to find total number of pixels = $(1024 \times 1080) = 1\,105\,920$ pixels
- 2 Now multiply number of pixels by colour depth then divide by 8 to give the number of bytes = $1\,105\,920 \times 32 = 35\,389\,440/8$ bytes = $4\,423\,680$ bytes
- 3 $64 \text{ GiB} = 64 \times 1024 \times 1024 \times 1024 = 68\,719\,476\,736$ bytes
- 4 Finally divide the memory stick size by the files size = $68\,719\,476\,736/4\,423\,680 = 15534$ photos.

EXAMPLE 3:

A camera detector has an array of 2048 by 2048 pixels and uses a colour depth of 16. Find the size of an image taken by this camera in MiB.

- 1 Multiply number of pixels in vertical and horizontal directions to find total number of pixels = $(2048 \times 2048) = 4\,194\,304$ pixels
- 2 Now multiply number of pixels by colour depth = $4\,194\,304 \times 16 = 67\,108\,864$ bits
- 3 Now divide number of bits by 8 to find the number of bytes in the file = $(67\,108\,864)/8 = 8\,388\,608$ bytes
- 4 Now divide by 1024×1024 to convert to MiB = $(8\,388\,608)/(1\,048\,576) = 8 \text{ MiB}$.

CALCULATING BIT-MAP IMAGE FILE SIZES

EXAMPLE 4:

An audio CD has a sample rate of 44 100 and a sample resolution of 16 bits. The music being sampled uses two channels to allow for stereo recording. Calculate the file size for a 60-minute recording.

1 Size of file =

sample rate (in Hz) × sample resolution (in bits) × length of sample (in seconds)

2 Size of sample = $(44\,100 \times 16 \times (60 \times 60)) = 2\,540\,160\,000$ bits

3 Multiply by 2 since there are two channels being used = $5\,080\,320\,000$ bits

4 Divide by 8 to find number of bytes = $(5\,080\,320\,000)/8 = 635\,040\,000$

5 Divide by 1024×1024 to convert to MiB = $635\,040\,000 / 1\,048\,576 = 605$ MiB.

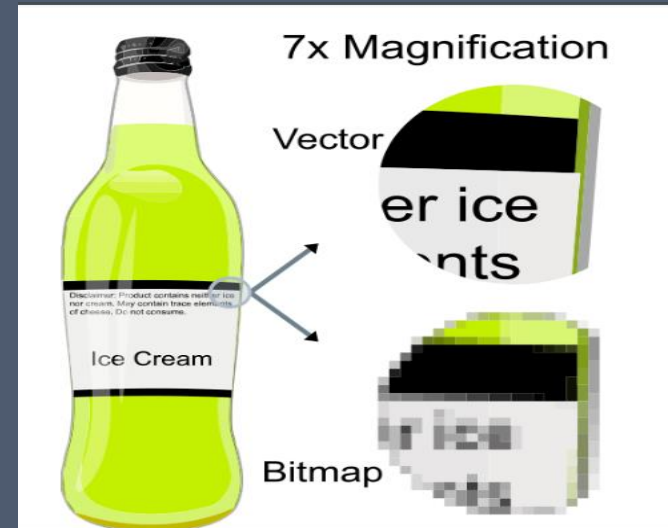
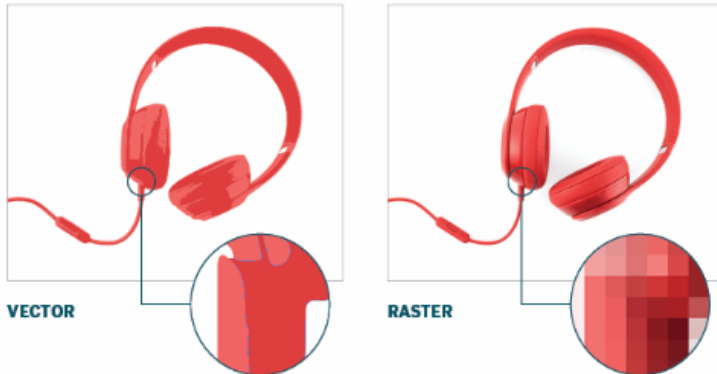
1.2 Text, sound and images:

Images:

VECTOR GRAPHICS ARE COMPUTER IMAGES CREATED USING A SEQUENCE OF COMMANDS OR MATHEMATICAL STATEMENTS THAT PLACE LINES AND SHAPES IN A TWO-DIMENSIONAL OR THREE-DIMENSIONAL SPACE. IN VECTOR GRAPHICS, A GRAPHIC ARTIST'S WORK, OR FILE, IS CREATED AND SAVED AS A SEQUENCE OF VECTOR STATEMENTS.

A VECTOR GRAPHIC FILE DESCRIBES A SERIES OF POINTS TO BE CONNECTED. THESE FILES ARE SOMETIMES CALLED GEOMETRIC FILES. IMAGES CREATED WITH TOOLS SUCH AS ADOBE ILLUSTRATOR AND COREL'S COREL DRAW ARE USUALLY VECTOR IMAGE FILES. SIMPLIFIED, VECTOR GRAPHICS ARE LIKE CONNECT-THE-DOTS DRAWINGS.

Vector vs. raster file



1.2 Text, sound and images:

Images:

Comparison between vector graphics and bit-map images

Vector graphic images	Bit-map images
made up of geometric shapes which require definition/attributes	made up of tiny pixels of different colours
to alter/edit the design, it is necessary to change each of the geometric shapes	possible to alter/edit each of the pixels to change the design of the image
they do not require large file size since it is made up of simple geometric shapes	because of the use of pixels (which give very accurate designs), the file size is very large
because the number of geometric shapes is limited, vector graphics are not usually very realistic	since images are built up pixel by pixel, the final image is usually very realistic
file formats are usually .svg, .cgm, .odg	file formats are usually .jpeg, .bmp, .png

1.1 DATA REPRESENTATION

1.3 DATA STORAGE AND COMPRESSION

By: Nouredine Tadjerout

Learning Objectives

1.3 Data storage and compression

1 Understand how data storage is measured

- Including:
 - bit
 - nibble
 - byte
 - kibibyte (KiB)
 - mebibyte (MiB)
 - gibibyte (GiB)
 - tebibyte (TiB)
 - pebibyte (PiB)
 - exbibyte (EiB)
- The amount of the previous denomination present in the data storage size, e.g.:
 - 8 bits in a byte
 - 1024 mebibytes in a gibibyte
- Answers must be given in the units specified in the question
- Information given may include:
 - image resolution and colour depth
 - sound sample rate, resolution and length of track
- Compression exists to reduce the size of the file
- The impact of this is, e.g.:
 - less bandwidth required
 - less storage space required
 - shorter transmission time

2 Calculate the file size of an image file and a sound file, using information given

3 Understand the purpose of and need for data compression

Learning Objectives



1.3 Data storage and compression

- 4 Understand how files are compressed using lossy and lossless compression methods
 - Lossless compression reduces the file size without permanent loss of data, e.g. run length encoding (RLE)
 - Lossy compression reduces the file size by permanently removing data, e.g. reducing resolution or colour depth, reducing sample rate or resolution

1.3 DATA STORAGE AND COMPRESSION

Mr. Nouredine Adjerout

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FILE FORMATS

IEC (INTERNATIONAL ELECTROTECHNICAL COMMISSION)

Name of memory size	Number of bytes	Equivalent denary value
1 kibibyte (1 KiB)	2^{10}	1024 bytes
1 mebibyte (1 MiB)	2^{20}	1048576 bytes
1 gibibyte (1 GiB)	2^{30}	1073741824 bytes
1 tebibyte (1 TiB)	2^{40}	1099511627776 bytes
1 pebibyte (1 PiB)	2^{50}	1125899906842624 bytes
1 exbibyte (1 EiB)	2^{60}	1152921504606846976 bytes

This system is more accurate. Internal memories (such as RAM and ROM) should be measured using the IEC system. A 64 GiB RAM could, therefore, store $64 \times 2_{30}$ bytes of data (68 719 476 736 bytes).

1.3 DATA STORAGE AND COMPRESSION

The Calculations In Section 1.3.2 Show That Sound And Image Files Can Be Very Large. It Is Therefore Necessary To Reduce (Or Compress) The Size Of A File For The Following Reasons:

- » To Save Storage Space On Devices Such As The Hard Disk Drive/Solid State Drive.
- » To Reduce The Time Taken To Stream A Music Or Video File.
- » To Reduce The Time Taken To Upload, Download Or Transfer A File Across A Network.
- » The Download/Upload Process Uses Up Network Bandwidth – This Is The maximum Rate Of Transfer Of Data Across A Network, Measured In Bits Per Second. This Occurs Whenever A File Is Downloaded, For Example, From A Server. Compressed Files Contain Fewer Bits Of Data Than Uncompressed Files And Therefore Use Less Bandwidth, Which Results In A Faster Data Transfer Rate.
- » Reduced File Size Also Reduces Costs. For Example, When Using Cloud Storage, the Cost Is Based On The Size Of The Files Stored. Also An Internet Service Provider (ISP) May Charge A User Based On The Amount Of Data Downloaded.

1.3 Data storage and compression

Lossy and loss less file compression

Click the link below to watch the video for compression:

<https://www.youtube.com/watch?v=29kUfFIIVJA>



1.3 Data storage and compression

Lossy and loss less file compression



Why we use compression

Compression is used to reduce the size of a file so it requires less:

- Bandwidth
- Storage space
- Transmission time

Lossy compression:

- Some data is lost and cannot be recovered.
- Greatly reduces file size.
- Reduces the quality of images/sound.
- Suitable for images, sound and video.
- Cannot be used with text or executable files.

Lossless compression:

- No data is lost, just encoded differently.
- Files are recovered to their original state.
- Can be used with all types of data.
- Less effective at reducing file size.
- Most suitable for text documents and executable files.

1.3 Data storage and compression

Lossy and loss less file compression

Click the link below to watch the video for Lossy and lossless compression methods:

https://www.youtube.com/watch?v=KJljOO48_vs



1.3 Data storage and compression

Lossy and loss less file compression



Lossy compression

Compression algorithm eliminates unnecessary data from the file

The original file cannot be reconstructed once it has been compressed

Information discarding
Perceptual music shaping

JPEG, MP3, MP4, WMV, MPG

Lossless compression

All the data from the original uncompressed file can be reconstructed

Important for files where any loss of data would be an issue

Run-length encoding (RLE)
Huffman coding

TIF, PDF, GIF, PNG, ZIP

1.3 Data storage and compression



Lossy and lossless file compression

File compression can either be **lossless** or **lossy**.

Lossy file compression:

With this technique, the file compression algorithm eliminates unnecessary data from the file. This means the original file cannot be reconstructed once it has been compressed. Lossy file compression results in some loss of detail when compared to the original file. The algorithms used in the lossy technique have to decide which parts of the file need to be retained and which parts can be discarded.

Example: When applying a lossy file compression algorithm to:

- » an image, it may reduce the resolution and/or the bit/colour depth
- » a sound file, it may reduce the sampling rate and/or the resolution.

Lossy files are smaller than lossless files which is of great benefit when considering storage and data transfer rate requirements. Common lossy file compression algorithms are:

- » MPEG-3 (MP3) and MPEG-4 (MP4)
- » JPEG.

1.3 Data storage and compression

Lossless and lossy file compression

Lossless file compression

With **LOSSLESS FILE COMPRESSION**, all the data bits from the original file are reconstructed when the file is again uncompressed. This is particularly important for files where loss of any data would be disastrous – for example, a spreadsheet file.

Lossy file compression

LOSSY FILE COMPRESSION is very different to lossless file compression. With this technique, the file compression algorithm eliminates unnecessary bits of data as seen in MP3 and jpeg formats.

It is impossible to get the original file back once it is compressed. This is why it is chosen for files where removing certain bits doesn't detract from the quality.

1.3 Data storage and compression

- A number of different file formats are used in computer systems. We will look at the following ones:
 - Musical Instrument Digital Interface (MIDI)
 - MP3
 - MP4
 - jpeg
 - text and number format.

<u>Sound</u>	<u>Pictures</u>	<u>Video</u>	<u>Text</u>	<u>Numbers</u>
• MP3	• JPEG	• MP4	• ASCII	• Real
• WAV	• GIF	• AVI	• TXT	• Integer
• AIFF	• TIFF	• WebM	• RTF	• Currency
• M4A	• BMP	• WMV	• DOC	• Date

1.3 Data storage and compression



MPEG-3 (MP3) and MPEG-4 (MP4)

<https://www.youtube.com/watch?v=BAAVCPIJuhY>



Noureddine Tadjerout

1.3 Data storage and compressio



MPEG-3 (MP3) and MPEG-4 (MP4)

MPEG-3 (MP3) uses technology known as **AUDIO COMPRESSION** to convert music and other sounds into an MP3 file format. Essentially, this compression technology will reduce the size of a normal music file by about 90 per cent. For example, an 80 megabyte music CD can be reduced to 8 megabytes using MP3 technology.

MP3 files are used in MP3 players, computers or mobile phones.

How can the original music file be reduced by 90 per cent whilst still retaining most of the music quality? This is done using file compression algorithms which use **PERCEPTUAL MUSIC SHAPING**; this essentially removes sounds that the human ear can't hear properly. For example, if two sounds are played at the same time, only the louder one can be heard by the ear, so the softer sound is eliminated.

1.3 Data storage and compressio



MPEG-3 (MP3) and MPEG-4 (MP4)

MP3 files use what is known as a **LOSSY FORMAT**, this means that the original file can't be put back together again. However, even the quality of MP3 files can be different since it depends on the **BIT RATE** – this is the number of bits per second used when creating the file. Bit rates are roughly between 80 and 320 kilobits per second; usually 200 or higher gives a sound quality close to a normal CD.

MPEG-4 (MP4) files are slightly different to MP3 files. This format allows the storage of multimedia files rather than just sound. Music, videos, photos and animation can all be stored in the MP4 format. Videos, for example, could be streamed over the internet using the MP4 format without losing any real discernable quality.

1.3 Data storage and compression



MPEG-3 (MP3) and MPEG-4 (MP4)

<https://www.youtube.com/watch?v=MmecPiKCIHk>

A screenshot of a video player. The video has a black background. In the top left, the text "TECH TALK" is written in blue. In the center, there is a screenshot of an audio waveform analysis software showing two waveforms. Below the waveforms, the text "Hear The Difference!" is written in white. To the right of the waveform screenshot, the text "Quality Loss with MP3" is written in large, yellow, italicized font. At the bottom left, the text "Double A" is written in white, italicized font.

1.3 Data storage and compression



MPEG-3 (MP3) and MPEG-4 (MP4)

The following information refers to a music track being recorded on a CD:

- music is sampled 44 100 times per second
 - each sample is 16 bits
 - each track requires sampling for left and right speakers
- (i) Calculate the number of bytes required to store one second of sampled music.
Show your working.

$$\frac{44100 \times 16 \times 2}{8}$$

176 400 bytes

1.3 Data storage and compression



MPEG-3 (MP3) and MPEG-4 (MP4)

(ii) A particular track is four minutes long.

Describe how you would calculate the number of megabytes required to store this track.

Allow follow through from part (i) on 176 400

$$\frac{4 \times 60 \times 176400}{1024 \times 1024}$$

1.3 Data storage and compression



MPEG-3 (MP3) and MPEG-4 (MP4)

- (c) When storing music tracks in a computer, the MP3 format is often used. This reduces file size by about 90%.

Explain how the music quality is apparently retained.

(c) any three from:

- mp3 is a lossy compressed format
- uses psycho-acoustic modelling
- and perceptual music/noise shaping
- certain parts of the music can be eliminated without significantly degrading the listener's experience
- removes sound that the human ear can't hear
- only keeps sounds human ear can hear better than others
- discards softer sound if two sounds played together

1.3 Data storage and compression

Joint Photographic Experts Group (jpeg) files)

<https://www.youtube.com/watch?v=Ba89cl9elg8>



1.3 Data storage and compression

Joint Photographic Experts Group (jpeg) files

Look at the following five photographs of the same car wheel:



The resolution of the photographs is reduced from A to E. Photographs A and B are very sharp whilst photograph D is very fuzzy and E is almost unrecognizable. This is the result of changing the number of **PIXELS** per centimeter used to store the image (that is, reducing the **PICTURE RESOLUTION**).

When a photographic file undergoes file compression, the size of the file is reduced. When file size is reduced then quality of the image will reduce. One of the file formats used to reduce photographic file sizes is known as **JPEG**.

1.3 Data storage and compression

Joint Photographic Experts Group (jpeg) files

A black and white image is 512 pixels by 256 pixels.

Calculate the file size of this image in kilobytes (KB) (1 KB = 1024 bytes).

$$\frac{512 \times 256}{8 \times 1024} = 16 \text{ KB}$$

Give a reason why it is important to estimate the file size of an image.

so it is possible to estimate how many images can be stored / to decide if it can be sent as an email attachment

1.3 Data storage and compression

Joint Photographic Experts Group (jpeg) files

This is another example of lossy file compression. Jpeg will reduce the RAW BITMAP image by a factor of between 5 and 15 depending on the quality of the original.

An image that is 2048 pixels wide and 1536 pixels high is equal to 2048×1536 pixels; in other words, 3 145 728 pixels. This is often referred to as a 3-megapixel image (although it is obviously slightly larger). A raw bitmap can often be referred to as a TIFF or BMP image (file extension .TIF or .BMP). The file size of this image is determined by the number of pixels. In the previous example, a 3-megapixel image would be 3 megapixels $\times$ 3 colours. In other words, 9 megabytes (each pixel occupies 3 bytes because it is made up of the three main colours: red, green and blue). TIFF and BMP are the highest image quality because, unlike jpeg, they are not in a compressed format.

The same image stored in jpeg format would probably occupy between 0.6 megabytes and 1.8 megabytes.

1.3 Data storage and compression



Lossless file compression

With this technique, all the data from the original uncompressed file can be reconstructed. This is particularly important for files where any loss of data would be disastrous (e.g. when transferring a large and complex spreadsheet or when downloading a large computer application).

Lossless file compression is designed so that none of the original detail from the file is lost.

1.3 Data storage and compression

Lossless and lossy file compression

Explain the difference between *lossless* and *lossy* data compression techniques.

- lossless designed to lose none of the original detail/lossless allows original file to be recreated exactly
- lossless technique based on some form of replacement
- mention of type of replacement, for example RLE, FLAC etc.
- by example: e.g. 000–1111–222222–333 = 3–0, 4–1, 6–2, 3–3 etc.
- maximum compression about 50%
- lossy may result in loss of detail compared to original file/lossy does not allow original file to be re-created exactly
- lossy techniques make decision about what parts of sound/sound file are important and discards other information
- only keeps sounds human ear can process/discards sounds most people cannot hear
- ... then applies lossless technique, for further reduction
- lossy compression can reduce to about 10%
- an example of jpeg, mp3 or other correct examples of compressed formats.

1.3 Data storage and compression

Key definitions

MPEG-3 (MP3)	Uses audio compression to store music in an MP3 file format
MPEG-4 (MP4)	Can store multimedia rather than just music
Joint Photographic Experts Group (JPEG)	JPEG files use a LOSSY format file compression method. JPEG is used to store photographs as a reduced file size
Lossless	All the data from the original files are reconstructed when the file is again uncompressed
Lossy	Unnecessary data is eliminated forming a file that can't be reconstructed to get back to the original file

1.3 Data storage and compression

IGCSE past paper Exam Question:1

A logo is designed as a bitmap image.

(a) Describe what is meant by a **bitmap image**.

.....

.....

.....

.....[2]

1.3 Data storage and compression

IGCSE past paper Exam Question:1

(b) A black and white bitmap image is shown.



(i) Explain how a computer can store this bitmap image.

.....

.....

.....

.....[2]

1.3 Data storage and compression

IGCSE past paper Exam Question:1

(ii) The image is compressed before it is attached to an email.

Explain how run-length encoding (RLE) will compress the image.

.....

.....

.....

.....[2]

(c) The finished logo is 500 pixels by 1000 pixels and uses 35 different colours.

Estimate the file size for the logo. Give your answer in kilobytes. Show your working.

Working

.....

.....

.....

.....

Answer[4]

1.3 Data storage and compression

IGCSE past paper Exam Answer:1

Question	Answer	Marks
2(a)	1 mark per bullet, max 2 ☐ Made up of pixels ☐ Each pixel has one colour ☐ Colour of each pixel stored as a binary number	2
2(b)(i)	1 mark per bullet, max 2 ☐ Each pixel requires only one bit (as there are only two colours) ☐ Black represented by 1 and white by 0 (or vice versa) ☐ Bits are stored for each pixel in sequence ☐ 11111 01010 01010 01010 01010	2
2(b)(ii)	1 mark for the explanation ☐ Stores the colour and the number of times it occurs 1 mark for example from ☐ An example from the bitmap given e.g. B5, W1, B1 and so on	2
2(c)	1 mark per bullet ☐ Number of pixels $500 \times 1000 (= 500\,000)$ ☐ 35 colours require 6 bits per pixel ☐ Number of bytes $(500\,000 \times 6) / 8 = 3\,000\,000 / 8 (= 375\,000)$ ☐ = 375 Kb	4

1.3 Data storage and compression

IGCSE past paper Exam Question:2

A recording of a concert is stored as a file. The file is compressed using lossy compression before it is streamed to users.

(i) State why this file needs to be compressed.

.....
..... [1]

(ii) Define the term **lossy compression**.

.....
..... [1]

(iii) The file could be compressed using lossless compression.

Explain why lossy compression is a more appropriate compression technique than lossless for this file.

.....

1.3 Data storage and compression

IGCSE past paper Answer:2

1(d)(i)	1 mark per bullet point to max 1 ☐ The data files are very large ☐ It would take a long time to send the uncompressed file // Compressed files will download faster ☐ A higher bandwidth would be needed to transmit the uncompressed file	1
1(d)(ii)	1 mark per bullet point to max 1 ☐ Data is lost ☐ The decompressed file is not the same as the original	1
1(d)(iii)	1 mark per bullet point to max 3 ☐ Lossy creates a smaller file than lossless // lossy compresses further than lossless ☐ The recording of the concert is a large file size and needs <u>significant</u> reduction in size ☐ Lossy removes detail which can be lost without people noticing ☐ By example e.g. reduction in sound quality <u>will not be noticed</u>	3

1.3 DATA STORAGE AND COMPRESSION

RUN-LENGTH ENCODING (RLE)

Run-length encoding (rle) is a simple form of lossless data compression that runs on sequences with the same value occurring many consecutive times. it encodes the sequence to store only a single value and its count. for example, consider a screen containing plain black text on a solid white background.

What is rle used for?

rle is particularly well suited to **palette-based bitmap images such as computer icons**, and was a popular image compression method on early online services such as compuserve before the advent of more sophisticated formats such as gif

Example: the sequence of data is stored as a single value and count. for example, **for a minute of a scene filmed at a beach there would be similar colours on screen for the duration of the shot, such as the blues of the sky and the sea, and the yellows of the sand.**

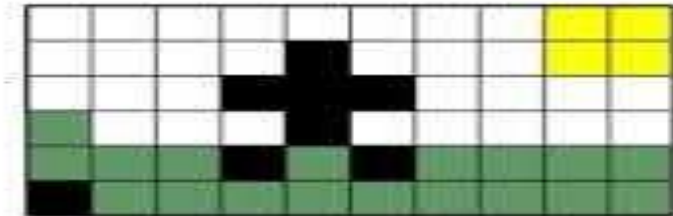
1.3 DATA STORAGE AND COMPRESSION

WATCH THE VIDEO FOR RUN-LENGTH ENCODING (RLE) (6 MINUTES)

[HTTPS://WWW.YOUTUBE.COM/WATCH?V=CAAESN8_ACU](https://www.youtube.com/watch?v=CAAESN8_ACU)

[HTTPS://WWW.CSFIELDGUIDE.ORG.NZ/EN/CHAPTERS/CODING-COMPRESSION/RUN-LENGTH-ENCODING/](https://www.csfieldguide.org.nz/en/chapters/coding-compression/run-length-encoding/)

How RLE works

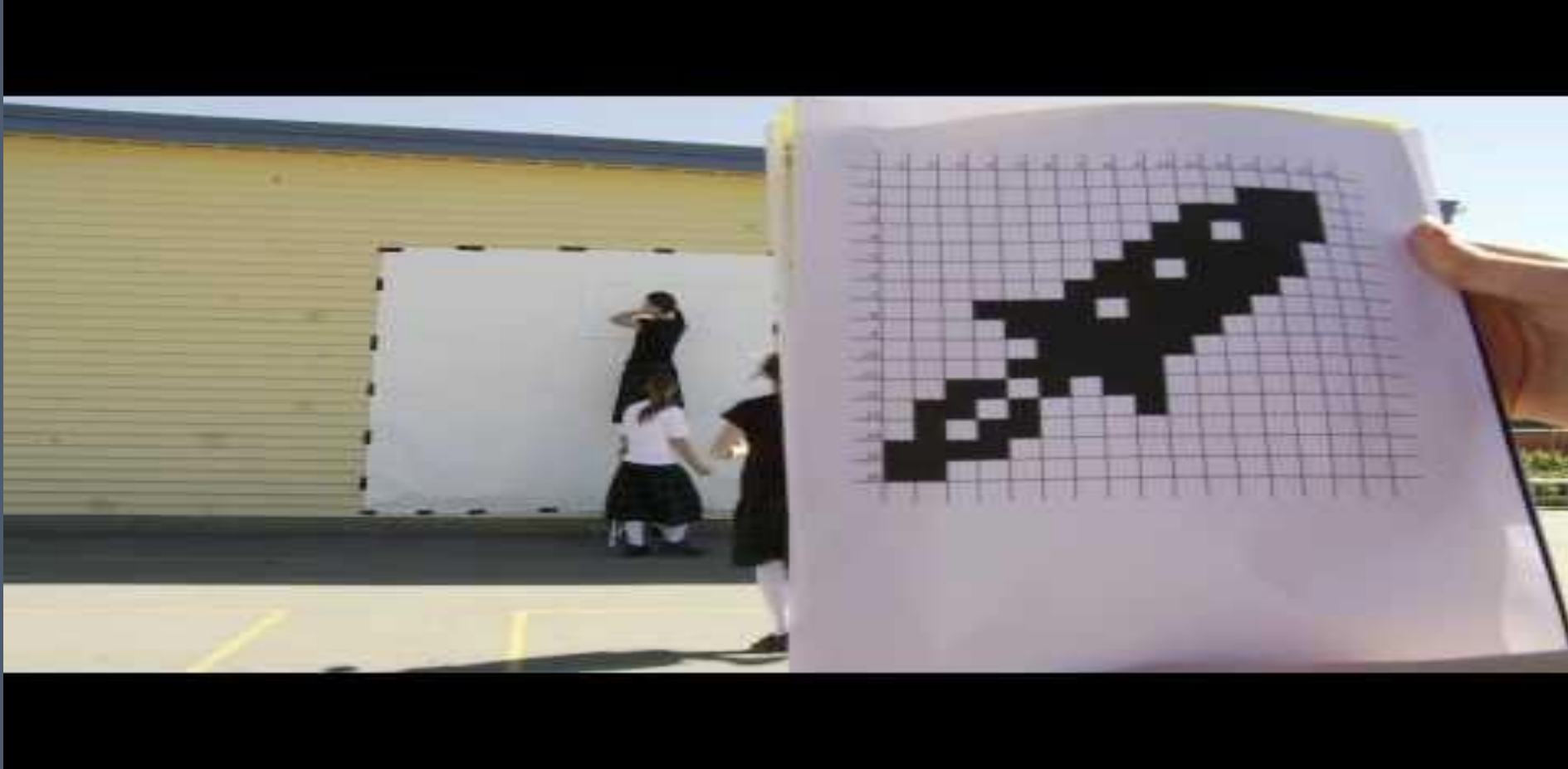


8	2		
4	1	3	2

1.3 DATA STORAGE AND COMPRESSION

WATCH THE VIDEO FOR RUN-LENGTH ENCODING (RLE) (6 MINUTES)

[HTTPS://YOUTU.BE/UAV2RUAJTJQ](https://youtu.be/UAV2RUAJTJQ)



1.3 Data storage and compression

Run-length encoding (RLE)

Run-length encoding (RLE) can be used for lossless compression of a number of different file formats:

- » it is a form of lossless/reversible file compression
- » it reduces the size of a string of adjacent, identical data (e.g. repeated colours in an image)
- » a repeating string is encoded into two values:
 - the first value represents the number of identical data items (e.g. characters) in the run
 - the second value represents the code of the data item (such as ASCII code if it is a keyboard character)
- » RLE is only effective where there is a long run of repeated units/bits.

Using RLE on text data

Consider the following text string: 'aaaaabbbbccddddd'. Assuming each character requires 1 byte then this string needs 16 bytes. If we assume ASCII code is being used, then the string can be coded as follows:

a	a	a	a	A	b	b	b	b	c	c	d	d	d	d	d
05 97					04 98				02 99		05 100				

1.3 Data storage and compression

Run-length encoding (RLE)

This means we have five characters with ASCII code 97, four characters with ASCII code 98, two characters with ASCII code 99 and five characters with ASCII code 100. Assuming each number in the second row requires 1 byte of memory, the RLE code will need 8 bytes. This is half the original file size.

One issue occurs with a string such as 'cdcdcdcdcd' where RLE compression isn't very effective. To cope with this, we use a flag. A flag preceding data indicates that what follows are the number of repeating units (for example, 255 05 97 where 255 is the flag and the other two numbers indicate that there are five items with ASCII code 97). When a flag is not used, the next byte(s) are taken with their face value and a run of 1 (for example, 01 99 means one character with ASCII code 99 follows).

Consider this example:

String	aaaaaaaa	bbbbbbbbbb	c	d	c	d	c	d	eeeeeeee
Code	08 97	10 98	01 99	01 100	01 99	01 100	01 99	01 100	08 101

1.3 Data storage and compression

Run-length encoding (RLE)

The original string contains 32 characters and would occupy 32 bytes of storage.

The coded version contains 18 values and would require 18 bytes of storage.

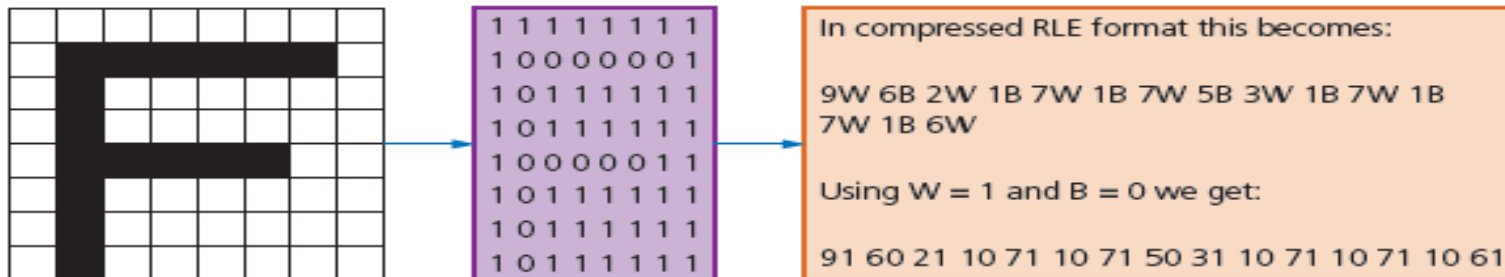
Introducing a flag (255 in this case) produces:

```
255 08 97 || 255 10 98 || 99 100 99 100 99 100 || 255 08 101
```

This has 15 values and would, therefore, require 15 bytes of storage. This is a reduction in file size of about 53% when compared to the original string.

? Example 1: Black and white image

Figure 1.12 shows the letter 'F' in a grid where each square requires 1 byte of storage. A white square has a value 1 and a black square a value of 0:



▲ **Figure 1.12** Using RLE with a black and white image

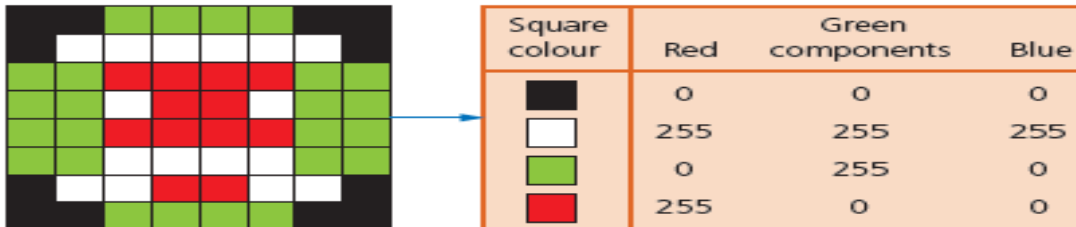
The 8 × 8 grid would need 64 bytes; the compressed RLE format has 30 values, and therefore needs only 30 bytes to store the image.

1.3 Data storage and compression

Run-length encoding (RLE)

? Example 2: Coloured images

Figure 1.13 shows an object in four colours. Each colour is made up of red, green and blue (RGB) according to the code on the right.



▲ Figure 1.13 Using RLE with a coloured image

This produces the following data: 2 0 0 0 4 0 255 0 3 0 0 0 6 255 255 255 1 0 0 0
2 0 255 0 4 255 0 4 0 255 0 1 255 255 255 2 255 0 0 1 255 255 255 4 0 255 0
4 255 0 0 4 0 255 0 4 255 255 255 2 0 255 0 1 0 0 0 2 255 255 255 2 255 0 0
2 255 255 255 3 0 0 0 4 0 255 0 2 0 0 0.

The original image (8×8 square) would need 3 bytes per square (to include all three RGB values). Therefore, the uncompressed file for this image is $8 \times 8 \times 3 = 192$ bytes.

The RLE code has 92 values, which means the compressed file will be 92 bytes in size. This gives a file reduction of about 52%. It should be noted that the file reductions in reality will not be as large as this due to other data which needs to be stored with the compressed file (e.g. a file header).

Question 1

- (c) When storing music tracks in a computer, the MP3 format is often used. This reduces file size by about 90%.

Explain how the music quality is apparently retained.

.....

.....

.....

.....

.....

.....[3]

Answer 1

(c) any three from:

- mp3 is a lossy compressed format
- uses psycho-acoustic modelling
- and perceptual music/noise shaping
- certain parts of the music can be eliminated without significantly degrading the listener's experience
- removes sound that the human ear can't hear
- only keeps sounds human ear can hear better than others
- discards softer sound if two sounds played together

[3]

Question 2

(b) The students use software to compress the sound clips before emailing them.

(i) Circle your chosen method of compression and justify your choice.

Lossy / Lossless

Justification:
.....
.....
.....[3]

Students also email images to the radio station for use on its website.

These are compressed before sending using run-length encoding (RLE).

(ii) Explain what is meant by run-length encoding.

.....
.....
.....
.....
.....
.....[3]

Question 2

(iii) The following diagrams show:

- the denary colour code that represents each colour
- the first three rows of a bitmap image

Colour symbol	Colour code (denary)
B	153
W	255

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	B	B	B	B	B	B	B	B	B	B	W	W	W	B	B	B
1	B	B	B	B	B	B	B	B	B	W	W	W	W	W	W	B
2	B	B	B	B	B	B	B	W	W	W	W	W	W	W	W	W
...																
95																

Show how RLE will compress the first three rows of this image.

Row 1:

Row 2:

Row 3:[2]

Exam: Answer 2

(b) (i) No mark awarded for identifying method. **Three marks** for justification. [3]

Lossy – Three points from:

- The human ear will not notice that the decompressed stream will not be identical to the original (file) / that parts of the original data have been discarded / removed / deleted.
- File size reduction is greater than using lossless.
- Email has limits on file sizes (on attachments) / a smaller file will take less time to transmit.
- The file may not need to be of high precision / accuracy.
- The producer has requested an mp3 file.

Lossless – Three points from:

- The file needs to be high precision / accuracy.
- None of the original data is lost / the decompressed file will be identical to the original.
- The producer has requested a flac file.

Answer 2

(ii) Three points from:

[3]

- Lossless method of compression.
- Reduces (the physical size of) a string of adjacent, identical characters/pixels / bytes etc..
- The repeating string (a run) is encoded into two values.
- One value represents the number of (identical) characters in the run (the run count).
- The other value is the code of the character / colour code of pixel etc. in the run (the run value).
- The run value and run count combination may be preceded by a control character.
- Any valid example given.

(iii) Two marks for three correct rows, one mark for two correct rows.

[2]

Row 1: 153 10 255 3 153 3

Row 2: 153 9 255 6 153 1

Row 3: 153 7 255 9

Alternative correct answer:

Row 1: 153 9 255 2 153 2

Row 2: 153 8 255 5 153 0

Row 3: 153 6 255 8

Question 3

(b) The computer also has bitmap software.

(i) Define the term **image resolution**.

.....
.....[1]

(ii) A picture is drawn and is saved as a 16-colour bitmap image.

State how many bits are used to encode the data for one pixel.

.....[1]

(iii) A second picture has width 8192 pixels and height 256 pixels. It is saved as a 256-colour bitmap.

Calculate the file size in kilobytes.

Show your working.

.....
.....
.....
.....
.....[3]

(iv) The actual bitmap file size will be larger than your calculated value as a bitmap file has a file header.

State **two** items of data that are stored in the file header.

1

2[2]

Answer 3

3(b)(i)	One from: ☐ The <u>number of pixels</u> per <u>unit measurement</u> 1 ☐ The number of pixels in an image 1 ☐ The number of pixels wide by the number of pixels high 1 ☐ Number of pixels per row by the number of rows 1	1
3(b)(ii)	4	1
3(b)(iii)	<i>Working: Max two from:</i> ☐ Number of pixels is 8192×256 1 ☐ One pixel will be stored as one byte 1 ☐ Number of kilobytes = $(8192 \times 256) / 1024$ 1 <i>Answer: One mark:</i> Number of kilobytes = 2048 KB 1	3
3(b)(iv)	Two from: ☐ Confirmation that the file is a BMP 1 ☐ File size 1 ☐ Location/offset of image data within the file 1 ☐ Dimensions of the image (in pixels) // image resolution 1 ☐ Colour depth (bits per pixel, 1, 4, 8, 16, 24 or 32) 1 ☐ Type of compression used, if any 1	Max 2

Question 4

A logo is designed as a bitmap image.

(a) Describe what is meant by a **bitmap image**.

.....

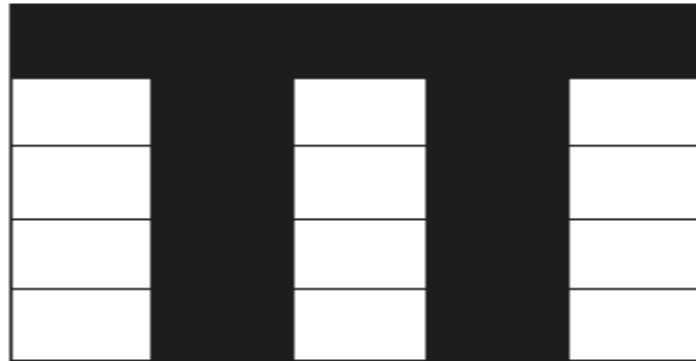
.....

.....

.....[2]

Question 4

(b) A black and white bitmap image is shown.



(i) Explain how a computer can store this bitmap image.

.....

.....

.....

.....[2]

(ii) The image is compressed before it is attached to an email.

Explain how run-length encoding (RLE) will compress the image.

.....

.....

.....

.....[2]

Question 4

- (c) The finished logo is 500 pixels by 1000 pixels and uses 35 different colours.

Estimate the file size for the logo. Give your answer in kilobytes. Show your working.

Working

.....

.....

.....

.....

Answer

[4]

- (d) The logo is redesigned as a vector graphic.

State **two** benefits of a vector graphic compared to a bitmap image. Give a reason for each benefit.

Benefit 1

.....

Reason 1

.....

Benefit 2

.....

Reason 2

.....

[4]

Answer 4

Question	Answer	Marks
2(a)	1 mark per bullet, max 2 ☐ Made up of pixels ☐ Each pixel has one colour ☐ Colour of each pixel stored as a binary number	2
2(b)(i)	1 mark per bullet, max 2 ☐ Each pixel requires only one bit (as there are only two colours) ☐ Black represented by 1 and white by 0 (or vice versa) ☐ Bits are stored for each pixel in sequence ☐ 11111 01010 01010 01010 01010	2
2(b)(ii)	1 mark for the explanation ☐ Stores the colour and the number of times it occurs 1 mark for example from ☐ An example from the bitmap given e.g. B5, W1, B1 and so on	2
2(c)	1 mark per bullet ☐ Number of pixels $500 \times 1000 (= 500\,000)$ ☐ 35 colours require 6 bits per pixel ☐ Number of bytes $(500\,000 \times 6) / 8 = 3\,000\,000 / 8 (= 375\,000)$ ☐ = 375 Kb	4
Question	Answer	Marks
2(d)	1 mark per bullet to max 2 marks per benefit ☐ Can resize it without pixilation ☐ Image is redrawn/recalculated with each adjustment ☐ Smaller file size ☐ Storing points/equations/commands etc., not individual pixels	4

Question 5

Xander creates a presentation that includes images, video and sound.

- (a) The images are bitmap images. A bitmap image can be made up of any number of colours. Each colour is represented by a unique binary number.

Draw **one** line from **each** box on the left, to the correct box on the right to identify the minimum number of bits needed to store each maximum number of colours.

Maximum number of colours

68

256

127

2

249

Minimum number of bits

1

2

3

7

8

9

Question 5

(b) One of the videos has a frame rate of 40 fps (frames per second).

(i) State what is meant by **40 fps**.

.....
..... [1]

(ii) One video uses interlaced encoding, and a second video uses progressive encoding.

Describe **two** differences between interlaced and progressive encoding.

1
.....
.....
.....
2
.....
.....
.....

[4]

Question 5

- (c) The sound track has a sampling rate of 88.2kHz and a sampling resolution of 32 bits.

State what is meant by a **sampling rate of 88.2kHz** and a **sampling resolution of 32 bits**.

Sampling rate of 88.2kHz

.....

.....

Sampling resolution of 32 bits

.....

.....

[2]

Answer 5

Question	Answer	Marks
5(a)	1 mark for each correctly linked box on the right Maximum number of colours Minimum number of bits 68 256 127 2 249 1 2 3 7 8 9	3

Answer 5

5(b)(i)	1 mark for correct answer 40 images or frames are <u>displayed/recorded</u> each second	1
5(b)(ii)	1 mark per bullet to max 4 ☐ Progressive – each frame contains (all the lines for) the complete image ☐ Interlaced – each frame contains half the (number of lines) of the complete image ☐ Progressive – frames are not divided into fields ☐ Interlaced – each frame divided into two fields // One field contains the data for the even numbered rows / lines and the other has the data for the odd numbered rows / lines. ☐ Progressive – complete frames are displayed in sequence ☐ Interlaced – data from two frames is displayed simultaneously ☐ Progressive – the number of images stored is the same as the frame rate // the rate of picture display is the same as the frame rate. ☐ Interlaced – the rate of picture display (the field rate) is twice the rate of image frame display (the frame rate). ☐ Progressive means the entire frame is refreshed each time ☐ Interlaced means only half the frame is refreshed ☐ Progressive has high bandwidth requirements ☐ Interlaced halves the transmission bandwidth requirements.	4

Answer 5

Question	Answer	Marks
5(c)	1 mark per bullet 88.2 kHz ☐ The sound wave is sampled <u>88200</u> times per second 32 bits ☐ Each sample is stored as a 32-bit binary number	2

Question 6

- 3 A touch screen has three squares where a selection can be made:



- (a) The x-coordinate of the centre of the three squares is held in three memory locations:

	Address	Memory contents
S	40	0000 1011 0100
T	41	0010 0101 0100
U	42	0100 0110 1100

- (i) Give the hexadecimal value of the memory contents for U.

.....
.....[1]

- (ii) Convert the denary number 40 into binary.

.....
.....[1]

Question 6

(b) Bitmap graphics are used to represent squares S, T and U.

These can be saved in a number of different image resolutions.

(i) Give the number of bits required to store each pixel for a black and white bitmap.

.....[1]

(ii) Identify how many bits are required to store each pixel for a 256-colour bitmap.

Explain your answer.

.....
.....
.....[2]

Answer 6

3 (a) (i) 46 C [1]

(ii) 101000 [1]

(b) (i) 1 bit [1]

(ii) – 8 bits are needed
– Each colour is represented by one of 256 values
– values 0 to 255/0000 0000 to 1111 1111
– $256 = 2^8$ [2]

Question 7

- 4 (a) Sound can be represented digitally in a computer.

Explain the terms sampling resolution and sampling rate.

Sampling resolution

.....

.....

.....

Sampling rate

.....

.....

.....[4]

Question 7

(b) The following information refers to a music track being recorded on a CD:

- music is sampled 44 100 times per second
- each sample is 16 bits
- each track requires sampling for left and right speakers

(i) Calculate the number of bytes required to store one second of sampled music.
Show your working.

.....
.....
.....
.....[2]

(ii) A particular track is four minutes long.

Describe how you would calculate the number of megabytes required to store this track.

.....
.....
.....
.....[2]

Question 7

- (c) When storing music tracks in a computer, the MP3 format is often used. This reduces file size by about 90%.

Explain how the music quality is apparently retained.

.....

.....

.....

.....

.....

.....[3]

Answer 7

4 (a) Sampling resolution (two marks)

- representation used to write samples in digital sound recording
- resolution is the number of distinct values available to encode/represent each sample
- specified by the number of bits used to store/record each sample
- sometimes referred to as bit depth
- the higher the sampling resolution the smaller the quantization error
- a higher sampling resolution results in less distortion of the sound
- usually 8 bit, 16 bit, 24 bit or 32 bit

Sampling rate (two marks)

- number of times that the amplitude of (analogue) sound wave is taken/measured
- per unit time/per second
- higher sampling rate results in more accurate digital representation

[4]

Answer 7

(b) (i) **one mark** for correct calculation, **one mark** for the answer

$$\frac{44100 \times 16 \times 2}{8} \quad (1 \text{ mark})$$

176 400 bytes (1 mark)

[2]

(ii) Allow follow through from part (i) on 176 400

$$\frac{4 \times 60 \times 176400}{1024 \times 1024} \quad \begin{array}{l} \text{one mark for numerator} \\ \text{one mark for denominator} \end{array} \quad [2]$$

(c) any **three** from:

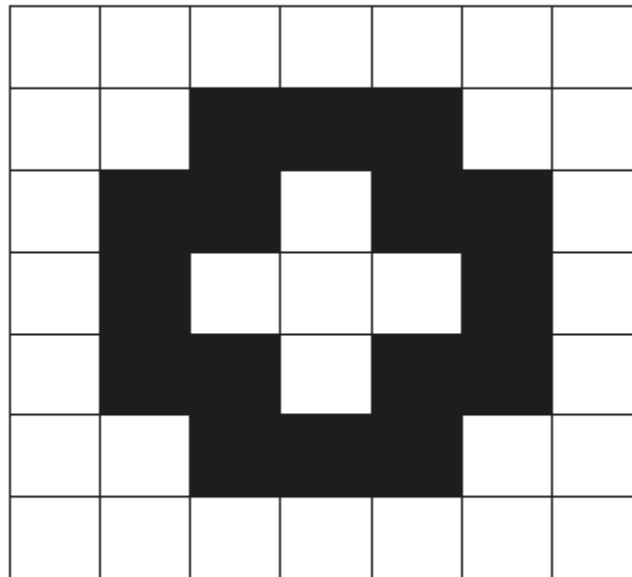
- mp3 is a lossy compressed format
- uses psycho-acoustic modelling
- and perceptual music/noise shaping
- certain parts of the music can be eliminated without significantly degrading the listener's experience
- removes sound that the human ear can't hear
- only keeps sounds human ear can hear better than others
- discards softer sound if two sounds played together

[3]

Question 8

12

- 6 A black and white bitmap image is shown.



- (a) State the **minimum** number of bits needed to represent each pixel in this image.

.....[1]

Question 8

- (b) Run-length encoding (RLE) is used to store the image with the following colour codes.

Colour	Code
Black	1A
White	3B

Show how run-length encoding is used to store the image.

.....
.....
.....[3]

- (c) An image has 30 different colours.

State the **minimum** number of bits needed to represent each pixel in the 30-colour image.

.....[1]

Question 8

(d) When the image is saved, a header is added to the file.

State the purpose of the **file header**. Give **two** examples of the file header contents.

Purpose

.....

Example 1

.....

Example 2

.....

[3]

Question 8

- (e) Graphics software is used to edit a digital photograph.

Give **three** features of graphics software that can be used to edit the photograph.

Describe the effect each has on the photograph.

Feature 1

Effect

.....

.....

Feature 2

Effect

.....

.....

Feature 3

Effect

.....

.....

Answer 8

Question	Answer	Marks
6(a)	1	1
6(b)	1 mark for correct method (colour code and number of pixels) 1 mark for first 7 groups correct 1 mark for remainder correct ☐ 3B9 1A3 3B3 1A2 3B1 1A2 3B2 ☐ 1A1 3B3 1A1 3B2 1A2 3B1 1A2 3B3 1A3 3B9	3
6(c)	5	1

Answer 8

Question	Answer	Marks
6(d)	1 mark for purpose ☐ Stores data about the file contents/image/metadata Max 2 marks for examples of contents ☐ <u>Confirmation</u> that the file is a BMP // confirmation of file type ☐ File size ☐ Location / offset of image data within the file ☐ Dimensions of the image (in pixels) // <u>image</u> resolution ☐ Colour depth (bits per pixel, 1, 4, 8, 16, 24 or 32) ☐ Type of compression used (if any)	3
6(e)	1 mark for naming tool, 1 mark for describing effect on the photograph e.g. ☐ Resize ☐ Increase / decrease the size of the image ☐ Crop ☐ Remove part of the image ☐ Blur ☐ Reduce the focus ☐ Red eye reduction ☐ Reduces red (light reflected from human eyes)	6

Question 9

1 A company is designing a website.

(a) The company creates a 4-colour bitmap image for the website as shown.

Each colour is represented by a letter, for example, G = grey, K = black.

G	R	G	K	W	R
G	R	G	K	W	R
G	R	G	K	W	R
G	R	G	K	W	R
G	G	G	K	K	R
W	W	W	W	K	R

(i) State the minimum number of bits needed to represent each pixel in the image in **part (a)**.

..... [1]

(ii) Calculate the minimum file size of the image shown in **part (a)**. Show your working.

Working

.....

.....

.....

File size

[3]

Question 9

- (b) The company takes a photograph of their office to put on the website. The photograph has a resolution of 1000 pixels by 1000 pixels. Two bytes per pixel are used to represent the colours.

- (i) Estimate the file size of the photograph in megabytes. Show your working.

Working

.....

.....

.....

Estimated file size

[4]

Question 9

- (ii) The file size of the photograph needs to be reduced before it is placed on the website.

Draw lines to link each method of reducing the file size of the image to:

- its description and
- its compression type, where appropriate.

Description	Method	Compression type
Removes pixels	Crop the photograph	Lossy
Reduces number of pixels per inch	Use run-length encoding	Lossless
Uses fewer bits per pixel	Use fewer colours	
Stores colour code and count of repetitions		

[5]

- (c) The company has created a logo for the website. The logo is a vector graphic.

Describe **two** reasons why a vector graphic is a sensible choice for the logo.

Reason 1

.....

.....

.....

Reason 2

.....

Answer 9

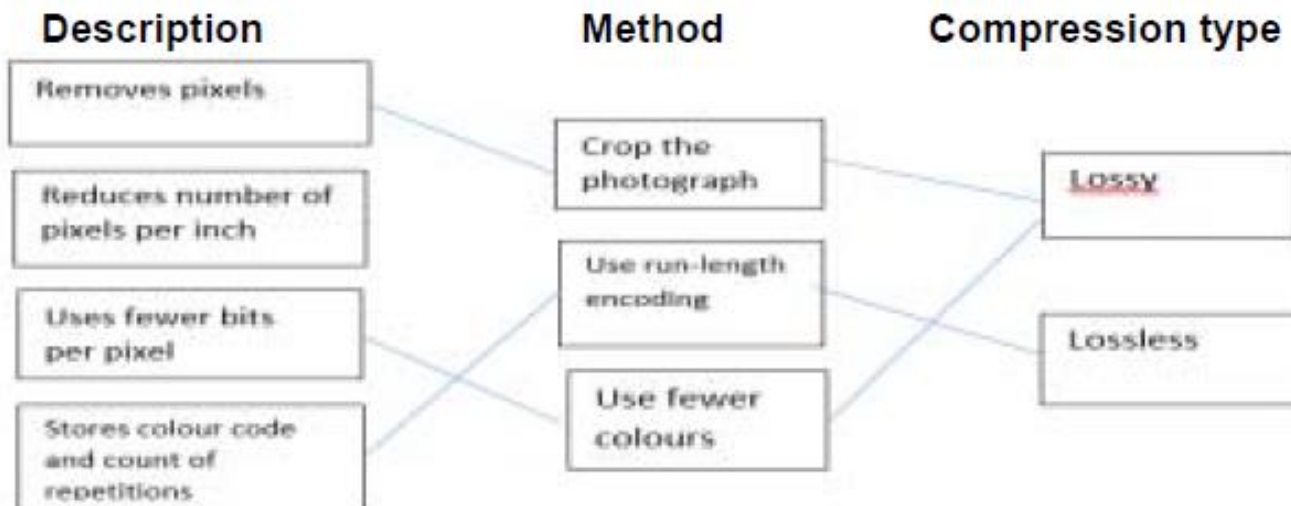
Question	Answer	Marks
1(a)(i)	2	1
1(a)(ii)	1 mark per bullet point • Number of pixels: $6 \times 6 // 36$ • Number of bits: Number of pixels (36) $\times 2 \dots$ • $\dots = 72$ <u>bits</u> // 9 <u>bytes</u>	3
1(b)(i)	1 mark per bullet point • Number of pixels: $1000 \times 1000 // 1\,000\,000$ • Number of bytes: Number of pixels (1 000 000) $\times 2 // 2\,000\,000 //$ Number of bits: Number of pixels (1 000 000) $\times 16 // 16\,000\,000$ • Conversion to <u>megabytes</u> • 2 (MB) // 1.91 (MB)	4

Answer 9

1(b)(ii)

1 mark per method correctly linked to its description max 3
1 mark for each compression type correctly linked to its method(s). max 2

5



1(c)

1 mark per bullet point. Max 2 marks for each reason.

4

- Smaller file size
- Can be transferred quicker/downloaded quicker
- Enlarges without pixilation
- Needs to be used on different screens / devices / resolutions

Question 10

1 A product designer is creating a poster.

(a) The designer creates a 6-colour bitmap image for the poster as shown.

Each colour is represented by a letter, for example, R = red, B = blue.

R	R	P	P	P	G
B	R	R	P	G	G
B	W	B	B	O	O
B	W	W	P	P	O
B	B	R	P	G	O
B	R	R	P	G	O

(i) State the minimum number of bits needed to represent each pixel in the image in **part (a)**.

.....[1]

Question 10

- (ii) Calculate the minimum file size of the image shown in **part (a)**. Show your working.

Working

.....

.....

File size

[3]

- (b) (i) The designer takes a photograph to put on the poster. The photograph has a resolution of 50 000 pixels by 50 000 pixels. The colours are represented using 4 bytes per pixel.

Estimate the file size of the photograph in gigabytes. Show your working.

Working

.....

.....

.....

Estimated file size

[4]

Question 10

- (ii) The photograph needs to be sent by email but the file size is too big. It needs to be compressed.

The table lists several methods of making an image file size smaller.

Tick (✓) **one** box on each row to indicate whether each method is lossy or lossless.

Compression method	Lossy	Lossless
Cropping the image		
Reducing the resolution of the image		
Using run-length encoding (RLE)		
Reducing the colour depth of the image		

[4]

- (c) Explain how run-length encoding would compress the image in **part (a)**.

.....

.....

.....

.....

Answer 10

Question	Answer	Marks
1(a)(i)	3	1
1(a)(ii)	1 mark per bullet point • Number of pixels: $6 \times 6 = 36$ • Number of bits: Number of pixels (36) $\times 3$ • 108 bits / 13.5 bytes	3
1(b)(i)	1 mark per bullet point • Number of pixels: $50\,000 \times 50\,000 = 2\,500\,000\,000$ • Number of bytes: Number of pixels (2 500 000 000) $\times \underline{4}$ // 10 000 000 000 // • Number of bits: Number of pixels (2 500 000 000) $\times \underline{32}$ // 80 000 000 000 • Conversion to gigabytes • 10 GB / 9.3 GB	4

Answer 10

1(b)(ii)	1 mark for correct tick in each row. <table><tr><th>Compression method</th><th>Lossy</th><th>Lossless</th></tr><tr><td>Cropping the image</td><td>✓</td><td></td></tr><tr><td>Reducing the resolution of the image</td><td>✓</td><td></td></tr><tr><td>Using run-length encoding (RLE)</td><td></td><td>✓</td></tr><tr><td>Reducing the colour depth of the image</td><td>✓</td><td></td></tr></table>	Compression method	Lossy	Lossless	Cropping the image	✓		Reducing the resolution of the image	✓		Using run-length encoding (RLE)		✓	Reducing the colour depth of the image	✓		4
Compression method	Lossy	Lossless															
Cropping the image	✓																
Reducing the resolution of the image	✓																
Using run-length encoding (RLE)		✓															
Reducing the colour depth of the image	✓																
1(c)	1 mark per bullet point to max 3 • Looks for runs of consecutive pixel of the same colour • Stores the colour value once and the number of times it occurs • Lossless method of compression • Reference to the given image in context	3															

Question 11

6 A student records a video using a digital camera.

(a) The recording uses interlaced encoding.

Describe **interlaced encoding**.

.....

.....

.....

.....

.....

.....

..... [3]

(b) State **one** benefit of using interlaced encoding compared to progressive encoding.

.....

..... [1]

Question 11

- (c) A video can be compressed using spatial redundancy or temporal redundancy.

Explain how **temporal redundancy** compresses a video.

.....

.....

.....

.....

..... [2]

- (d) A sound track is recorded for the video.

(i) Describe how a computer encodes the sound track.

.....

.....

.....

.....

.....

..... [3]

Question 11

- (ii) Explain how the sampling rate and sampling resolution affect the file size of the sound track.

Sampling rate

.....

Sampling resolution

.....

[2]

Answer 11

Question	Answer	Marks
6(a)	1 mark per bullet point to max 3 ☐ The data from a single frame is split into two separate fields ☐ One field has data for the odd numbered <u>rows/lines</u> and the other field has data for the even numbered <u>rows/lines</u> ☐ Odd numbered line fields alternate with even numbered line fields ☐ The viewer sees data from two frames simultaneously	3
6(b)	1 mark per bullet point to max 1 ☐ Produces what appears to be a higher refresh rate ☐ Lower bandwidth needed // Halves the bandwidth requirements	1
6(c)	1 mark per bullet point to max 2 ☐ Identifies pixels that do not change between frames ☐ Records only the differences between the frames	2

Answer 11

6(d)(i)	1 mark per bullet point to max 3 ☐ The amplitude of the wave is measured /sound wave is sampled ☐ At <u>set/regular</u> time intervals ☐ Each sample is stored as a binary number ☐ Samples are stored in order in a file	3
6(d)(ii)	1 mark per bullet point to max 1 for each Sample rate: ☐ Increasing the sample rate means more samples per second hence more bits per second <u>and</u> larger file size ☐ Decreasing the sample rate means fewer samples per second hence fewer bits per second <u>and</u> smaller file size Sample resolution: ☐ A higher sampling resolution means more bits per sample <u>and</u> a larger file size ☐ A lower sampling resolution means fewer bits per sample. a smaller file size	2

Question and Answer 12:

5 The following text is stored as a text file:

She sells sea shells on the seashore. The shells that she sells are sea shells I am sure.

Explain how lossless compression would compress this file.

.....

Five from:

- A (compression) algorithm is used
- No data is removed in the process // original file can be restored
- Repeated words (are identified) // Patterns in the data (are identified)
- ... and are indexed/put into a table // by example
- ... and are replaced with their index // by example
- ... and their positions are stored (in the table) // by example
- ... and the number of times the word/pattern appears is stored (in the table) // by example

NOTE: Other valid methods of lossless compression can be awarded marks

Question and Answer 13:

4 Michele wants to email a file to Elsa. The file is too large so it must be compressed.

(a) Name **two** types of compression that Michele could use.

Compression type 1 **Lossy (compression)**

Compression type 2 **Lossless (compression)**

[2]

(b) The file Michele is sending contains the source code for a large computer program.

Identify which type of compression would be most suitable for Michele to use.

Explain your choice.

Compression type..... **Lossless (compression)**

Explanation

Any **three** from:

- The file can be restored/decompressed to the exact same state it was before compression/ to original
- (It is a computer program so) no data can be lost // Lossy would remove data
- Will not run correctly (with any other compression)
- (Lossless) will give repeating words/sections of word a value// RLE is used // Other valid examples of methods of lossless compression
- Value is recorded in an index

[4]

Question and Answer 14:

(b) Nancy wants to email the photos to Nadia.

Many of the photos are very large files, so Nancy needs to reduce their file size as much as possible.

Identify which type of compression would be most suitable for Nancy to use. Explain your choice.

Compression type **Lossy**

Any **three** from:

- **Lossy** would reduce the file size **more** (than lossless)
- The **redundant** data can be removed from the files // by example (must be about redundant data)
- Images can still be a similar quality
- There is no requirement for the files to be exactly the same as original file
- Photos can be sent quicker // faster to upload // faster to download

(d) The table contains **four** different file formats that use compression.

Tick (✓) to show whether each file format uses **lossy** or **lossless** compression.

File format	Lossy (✓)	Lossless (✓)
.jpeg	✓	
.mp3	✓	
.mp4	✓	
.zip		✓

Question and Answer 15:

(c) Large files can be compressed to reduce their file size.

Two types of compression that can be used are lossy and lossless.

Explain how a file is compressed using lossless compression.

Three from:

- ☐ a compression algorithm is used
- ☐ no data is lost in the process
- ☐ repeated words/patterns can be indexed // repeated sections of words/patterns can be indexed // given by example
- ☐ The indexed words/patterns can be replaced with numerical values // given by example

(b) File compression is one example of a utility program.

Tick (✓) **one** box on each row to indicate whether the action is an example of lossy compression or lossless compression.

Action	Lossy	Lossless
Reducing the resolution of an image	✓	
Using run-length encoding on a text file		✓
Reducing the sampling rate of a sound file	✓	